A two-feedback Vicsek-Couzin flock: density-stratified heading correlations as the scale-invariant fingerprint of motility–noise coupling

Yaya Youssouf Yaya

Département de Mathématiques, École Normale Supérieure de N'Djaména, BP 206, N'Djaména, Tchad
Chercheur Associé, LMA, U.F.R. Sciences et Technologies, Université Assane Seck de Ziguinchor, BP 523, Diabir, Zigui
yy.y@zig.univ.sn ORCID: 0000-0003-0781-4923

May 25, 2026

Abstract

In the standard Vicsek model the self-propulsion speed and the angular-noise distribution are global constants. Two strands of the active-matter literature have softened each of them in turn. A density-dependent speed is enough to drive motility-induced phase separation, and heavy-tailed Lévy reorientations reshape the order–disorder transition; what neither strand has resolved is what happens when the two adaptations operate together. We close that gap by gating both the speed and the stability index of an α-stable angular kick through one shared sigmoid in the local neighbour count, so crowded particles are simultaneously slow and Gaussian-like while isolated ones are fast and Cauchy-like. A controlled seven-size finite-size scaling against the original Vicsek model and three single-channel ablations separates the two regimes cleanly. Only the motility-active modes develop a critical susceptibility scaling; the three fixed-noise-shape modes plateau with slopes statistically indistinguishable from zero. The full model is super-additive on this slope with respect to the motility-only ablation, and the synergy persists across all cumulative size budgets. The density-stratified heading correlation inside the dense phase, its dynamic counterpart $C(\tau)$, and the maximum-cluster size all converge on the same picture: the noise-shape feedback rectifies Cauchy bursts into Gaussian-like kicks inside the motility-driven dense phase, and the two channels cooperate super-additively. A decoupled-sigmoid sweep on a 25-cell threshold grid shows that the motility threshold alone controls the sign and amplitude of the rectification, while the noise-shape threshold is essentially passive.

Highlights

- We resolve a regime previously studied only in isolation — density-dependent motility and heavy-tailed Lévy reorientations — by activating both feedbacks in the same Vicsek–Couzin flock through a shared sigmoid.
- Heavy-tailed reorientations on their own do not drive a critical $\chi_{\max}(L)$ scaling on the seven-size lever; only the motility channel does, and the noise-shape adaptation adds super-additively on top of it.
- The density-stratified heading correlation inside the dense phase records the rectification with high precision ($\Delta g \simeq +0.15$, $z \simeq +14$ at $L = 30$); its dynamic counterpart $C(\tau)$ holds the same gap over four decades in lag.
- A 25-cell decoupled-sigmoid sweep shows the motility threshold $n_\star^v$ alone controls the sign and magnitude of the rectification ($R^2 = 0.88$ on $n_\star^v$, $R^2 = 0.04$ on the noise-shape threshold); the density-triggered slowdown is a necessary precondition for the heavy-tailed noise rectification to add coherence rather than destroy it.

1 Introduction

The diagnostic toolbox most often applied to self-propelled flocks reads the order parameter and its global fluctuations. The Vicsek model [Vicsek et al., 1995] keeps two ingredients of that flock — metric alignment with neighbours and an external angular noise — and freezes everything else: every particle moves at the same speed v_0, and every angular kick is drawn from the same Gaussian. Both choices are convenient. Both are physically wrong in interesting ways. Self-propulsion speed responds to local crowding in essentially every system one might want to model. Confluent epithelia jam and arrest [Bi et al., 2016], swarming bacteria switch between planktonic and biofilm regimes through quorum sensing, and traffic flows of vehicles, pedestrians or migrating animals all carry a marked speed–density anti-correlation [Greenshields, 1935, Lighthill and Whitham, 1955]. The reorientation statistics are no closer to Gaussian: heavy-tailed Lévy turns have been documented in foragers, T-cells and swarming bacteria [Reynolds, 2018, Harris et al., 2012, Ariel et al., 2015]. The biologically honest model therefore lets both inputs depend on the local environment, and the question we ask here is what such a double dependence does to the dense phase that the alignment interaction otherwise tries to build.

The two relaxations of the canonical model live in different parts of the active-matter literature. Cates and Tailleur [Cates and Tailleur, 2015] proved that a density-dependent speed $v(\rho)$ with $\partial \ln v/\partial \ln \rho < -1$ is enough to drive motility-induced phase separation (MIPS), in which a dense slow phase coexists with a dilute fast phase even with no alignment interaction at all. Heavy-tailed angular noise has been studied on a separate track and is known to reshape the order–disorder transition of metric Vicsek dynamics. To our knowledge, the regime in which both adaptations are switched on together has not been resolved. The question is not academic. The angular-noise distribution sets how particles escape spatial structures, while the density-dependent speed sets those structures in the first place, and the two channels therefore interrogate each other inside the same dense regions.

We tie the two adaptations together in the simplest way available. A single sigmoid in the local neighbour count n_i gates both the self-propulsion speed and the angular-noise stability index, so that a crowded particle is simultaneously slow and Gaussian and an isolated particle is simultaneously fast and Cauchy. Switching both channels off ($v_i \equiv v_0$, $\alpha_i \equiv 2$) recovers the original Vicsek dynamics, against which we benchmark at every step. Two single-channel ablations isolate the individual contributions: a motility-only mode with adaptive v_i and fixed Cauchy $\alpha = 1$, and a noise-shape-only mode with fixed $v_i = v_{\max}$ and adaptive $\alpha_i \in [1, 2]$.

A short quantitative preview helps place the results that follow. The finite-size scan covers $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$. The density-separation index s_{sep}, the standard MIPS diagnostic, plateaus at $\simeq 1.70$ at the largest sizes for both the motility-only ablation and the double-adaptive model, against 1.13–1.16 for the original Vicsek reference, the heavy-tailed Cauchy reference, and the noise-shape-only ablation: only the two motility-active modes produce spatial phase separation. A log–log fit of the susceptibility scaling $\chi_{\max} \sim L^a$ on the seven sizes gives $a_{\text{Vicsek}} = +0.01$, $a_{\text{Cauchy}} = -0.12$, $a_{\text{noise}} = +0.00$, $a_{\text{motility}} = +2.34$, and $a_{\text{full}} = +2.95$. The separation between modes is qualitative, not just quantitative. The three fixed-noise-shape modes (Vicsek–Gaussian, Cauchy reference, noise-shape-only ablation) all carry slopes statistically indistinguishable from zero, with the Cauchy reference robustly negative (95% bootstrap CI $[-0.15, -0.08]$); only the two motility-active modes develop a critical $\chi_{\max}(L)$ scaling. The synergy diagnostic $\Delta = (a_{\text{full}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}})$ would vanish for two channels combining linearly above the Cauchy backdrop. Instead it traces a persistently positive trajectory across the cumulative size budget: $\Delta_4 = +0.69$ on the four smallest sizes, $\Delta_5 = +1.45$ once $L = 64$ is folded in, $\Delta_6 = +0.46$ at $L = 90$, and $\Delta_7 = +0.48$ at $L = 128$. The Cauchy-only modes contribute essentially no susceptibility scaling on their own, so Δ is dominated by the difference between the full and motility-only slopes, $a_{\text{full}} - a_{\text{motility}} \simeq +0.6$, well outside the bootstrap envelope of zero on the four smallest sizes. The signal persists through seven sizes: the noise-shape feedback adds super-additively to the motility-driven critical scaling rather than washing out in the asymptotic regime.

The mechanism behind these numbers is direct. Inside the dense phase that the motility channel produces, the shared sigmoid pushes the local stability index toward $\alpha_{\max} = 2$, and the Cauchy bursts that would otherwise scramble local alignment are rectified into Gaussian-like kicks. The dense-quartile heading correlation records the rectification cleanly: at $L = 30$, a ten-seed measurement gives $g(r \simeq 0.6) = 0.542 \pm 0.007$ in the motility-only ablation against 0.695 ± 0.008 in the double-adaptive model, a paired gap of $+0.153 \pm 0.011$ ($z = +13.6$). The dilute-quartile correlation, by contrast, is essentially insensitive to the rectification and stays near zero in both modes; the noise-shape adaptation acts only where the shared sigmoid sends it, in the dense phase where particles are slow and crowded. The cluster diagnostics sharpen the geometric picture. At $L = 30$, the double-adaptive model produces 162.0 ± 14.5 dense clusters per snapshot against 187.9 ± 12.3 in the motility-only ablation, a paired difference of -25.9 ± 22.6 ($z = -1.15$, not significant); but the maximum-cluster size is markedly larger in the double-adaptive model, at 176.6 ± 42.3 particles versus 55.0 ± 7.5 in motility-only ($z = +2.7$). The rectification therefore fuses dense patches into a few large droplets rather than multiplying smaller ones. A two-dimensional decoupled-sigmoid sweep (Sec. 3.3.4) sharpens the picture by exposing the dominant control variable. On a 25-cell grid $(n_\star^v, n_\star^\alpha) \in \{1, 2, 3, 5, 8\}^2$ the sign and amplitude of the dense-quartile gap are governed almost entirely by the motility threshold $n_\star^v$: a linear regression captures $R^2 = 0.88$ on $n_\star^v$ alone with $\Delta g \simeq -0.144\, n_\star^v + 0.52$, while the analogous fit on the noise-shape threshold $n_\star^\alpha$ captures only $R^2 = 0.04$. The biological reading is unambiguous: the density-triggered slowdown is a necessary precondition; without it, the heavy-tailed noise rectification damages dense-phase coherence instead of building it.

2 Model and methods

2.1 Dynamics

We work with N point particles in a square box of side L with periodic boundary conditions. Particle i carries a position $\mathbf{x}_i \in [0, L)^2$ and a heading $\theta_i \in (-\pi, \pi]$, advanced synchronously through

$$\theta_i(t+1) = \theta_i^\star(t+1) + \xi_i(t), \qquad \mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_i(t+1)\,\vec{e}_i(t+1) \pmod{L}, \tag{1}$$

with $\vec{e}_i = (\cos\theta_i, \sin\theta_i)$. The desired heading $\theta_i^\star$ follows the Couzin priority rule [Couzin et al., 2002], made explicit by introducing the repulsion set $\mathcal{R}_i = \{j \neq i : |\mathbf{r}_{ij}| < R_r\}$ with $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, and the alignment set $\mathcal{A}_i = \{j \neq i : R_r \leq |\mathbf{r}_{ij}| < R_a\}$. Each particle senses omnidirectionally: every neighbour in the repulsion disc or the alignment annulus enters the corresponding set, regardless of its angular position relative to $\vec{e}_i(t)$. Repulsion takes priority over alignment, and alignment over inertia:

$$\theta_i^\star(t+1) = \begin{cases} \arg\Big(-\sum_{j\in\mathcal{R}_i} \mathbf{r}_{ij}/|\mathbf{r}_{ij}|\Big), & \mathcal{R}_i \neq \emptyset, \\ \arg\Big(\sum_{j\in\mathcal{A}_i} e^{i\theta_j(t)}\Big), & \mathcal{R}_i = \emptyset \text{ and } \mathcal{A}_i \neq \emptyset, \\ \theta_i(t), & \mathcal{R}_i = \mathcal{A}_i = \emptyset. \end{cases} \tag{2}$$

The three branches respectively send the particle away from the centroid of its repulsion-zone neighbours, align it with the mean heading of its alignment-zone neighbours, and preserve its current heading when no neighbour is present. Interactions run in $O(N)$ on a cell-list partition.

The angular kick ξ_i is a symmetric α-stable variate with the stability index set particle by particle,

$$\xi_i(t) \sim S_{\alpha_i(t)}\big(c = \eta,\ \beta_{\text{stab}} = 0,\ \mu = 0\big), \tag{3}$$

which we sample with the Chambers–Mallows–Stuck representation [Chambers et al., 1976] from a uniform U and an exponential W. The Gaussian limit $\alpha_i = 2$ is handled separately and recovers the original Vicsek noise.

2.2 The shared sigmoid

The single departure from canonical Vicsek dynamics is that both the propulsion speed and the noise stability index become local fields, both controlled by the same instantaneous neighbor count $n_i(t) = |\mathcal{A}_i(t)|$ through one sigmoid,

$$\sigma_i(t) = \frac{1}{1 + \exp\big[-s\,(n_i(t) - n_\star)\big]}, \tag{4}$$

$$v_i(t) = v_{\max} - (v_{\max} - v_{\min})\,\sigma_i(t), \tag{5}$$

$$\alpha_i(t) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\sigma_i(t). \tag{6}$$

A crowded particle ($n_i \gg n_\star$) sits at $\sigma_i \to 1$, hence at $v_i \to v_{\min}$ and $\alpha_i \to \alpha_{\max} = 2$. An isolated particle ($n_i \ll n_\star$) sits at $\sigma_i \to 0$, $v_i \to v_{\max}$, and $\alpha_i \to \alpha_{\min} = 1$. The two channels are locked: slow means Gaussian, fast means Cauchy. No operating point lets them fight each other locally.

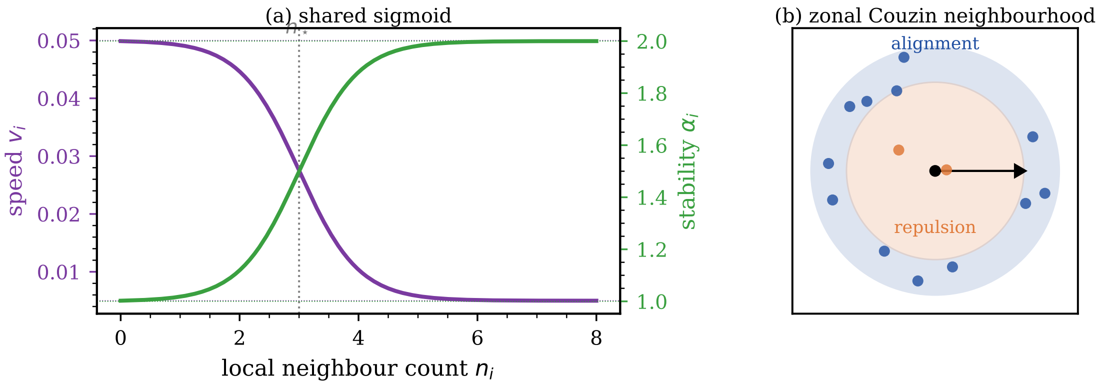

Figure 1. Construction of the two-feedback model. Panel (a): the shared sigmoid maps the local neighbor count n_i onto a propulsion speed v_i (purple) and a stability index α_i (green) that move together as the particle goes from isolated to crowded. Panel (b): the Couzin zonal neighbourhood used to compute n_i: an inner repulsion disc and an outer alignment annulus, sensed omnidirectionally without any angular masking.

2.3 Reference and ablation modes

The two-feedback model (henceforth *double-adaptive*) is benchmarked against three controls that share the same code path. Setting $v_{\min} = v_{\max}$ together with $\alpha_{\min} = \alpha_{\max} = 2$ recovers the original Vicsek dynamics (Gaussian noise, fixed speed); we call this the *Vicsek reference*. Setting instead $\alpha_{\min} = \alpha_{\max} = 1$ gives the heavy-tailed Cauchy analogue at the same scale, which we use as the *Cauchy reference* and treat as the backdrop on top of which the two feedbacks are switched on. Two single-channel ablations sit between these references and the full model: the *noise-shape only* mode activates the noise-shape channel alone with v_i frozen, while the *motility only* mode activates the motility channel alone with α_i frozen at the Cauchy value.

2.4 Order parameters and observables

The polar order parameter is the magnitude of the average heading,

$$\langle\varphi\rangle = \Big\langle \frac{1}{N}\big|\textstyle\sum_i e^{i\theta_i}\big|\Big\rangle, \tag{7}$$

the susceptibility is its scaled variance $\chi = N\,\mathrm{Var}(\varphi)$, and the Binder cumulant reads $U_4 = 1 - \langle\varphi^4\rangle/(3\langle\varphi^2\rangle^2)$. To diagnose spatial phase separation we use a coarse density-separation index

$$s_{\rm sep} \equiv \max_{\rm bins} \frac{n_{\rm bin}}{\mathrm{med}(n_{\rm bin})}, \tag{8}$$

the ratio of the densest grid cell to the median cell on a 10×10 partition. A homogeneous fluid sits at $s_{\rm sep} \simeq 1$; a phase-separated configuration sits markedly above. The 10×10 binning fixes a length scale, and we cross-checked the result against a grid-free pair-correlation analogue (the 95th-percentile / median ratio of the per-particle local-density field within radius $R = 1$). The two indices are correlated and rank the modes consistently on the snapshot configurations, so the grid choice does not bias the comparison; the grid-free implementation ships with the code. To tell a traveling-band soliton from an isotropic dense patch, we add the band-soliton index

$$b \equiv \frac{\max_{x_\parallel} \sigma(x_\parallel)}{\langle \sigma \rangle}, \tag{9}$$

the maximum of the time-averaged density profile along the instantaneous polar-flow axis $x_\parallel$ (read off the global polar-order vector at each measurement) divided by the spatial mean. A metric-Vicsek traveling band gives $b \gtrsim 1.5$; an isotropic patch that reorganises faster than its own polar axis gives $b \simeq 1$.

2.5 Numerical protocol

We run the seven sizes $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$, three seeds per (mode, L, η) cell, with 1500 warm-up and 1000 measurement steps from an aligned initial condition $\theta_i \equiv 0$. The noise grid is $\eta \in \{0.005, 0.010, 0.020, 0.035, 0.050, 0.075, 0.100, 0.150, 0.200, 0.300\}$. We fix $R_r = 0.5$, $R_a = 0.7$, sigmoid threshold $n_\star = 3$, sigmoid slope $s = 2$, and adaptation bounds $v_{\min} = 0.005$, $v_{\max} = 0.05$, $\alpha_{\min} = 1$, $\alpha_{\max} = 2$. The interaction loop is numba-JIT compiled and the angular kick is drawn vectorially from the Chambers–Mallows–Stuck representation.

A small-η robustness check refines the noise grid below $\eta = 0.005$ with $\{0.001, 0.002, 0.003, 0.0075\}$ at the first five sizes (two seeds), and confirms that $\chi_{\max}$ does not lurk outside the original window, with the single exception of the noise-shape-only ablation at $L = 45$, which we report with its updated value.

3 Results

Before turning to the quantitative diagnostics, we set a visual reference for what the two-feedback dynamics produces in real space at the working point used throughout the paper. We run the double-adaptive model at $L = 30$, $N = 2000$, after 3000 warm-up steps, at three contrasted noise levels spanning the ordered, near-critical, and disordered regimes; the configurations are shown in Fig. 2. At $\eta = 0.020$ the flock is almost fully aligned and dense. At the near-critical $\eta = 0.10$ a clearly inhomogeneous configuration emerges, with locally ordered slow patches embedded in a more disordered fast gas. At $\eta = 0.30$ the particles fill the box without a coherent global heading. The near-critical panel is the operating point at which the noise-shape adaptation acts on the Cauchy bursts of the dilute regions and at which the rectification mechanism documented below is active.

3.1 Global diagnostics

The first set of diagnostics reduces each configuration to a single scalar comparable across modes and sizes: the susceptibility scaling, the density-separation index, and the real-space configurations behind both. These are the standard MIPS and Vicsek tools.

3.1.1 Finite-size scaling of the susceptibility

The snapshots above set the scene at one box size; deciding whether the two adaptations leave a thermodynamic trace requires a controlled finite-size scan with a fixed protocol across modes. We therefore run all four heavy-tailed modes — fixed-Cauchy reference, noise-shape only, motility

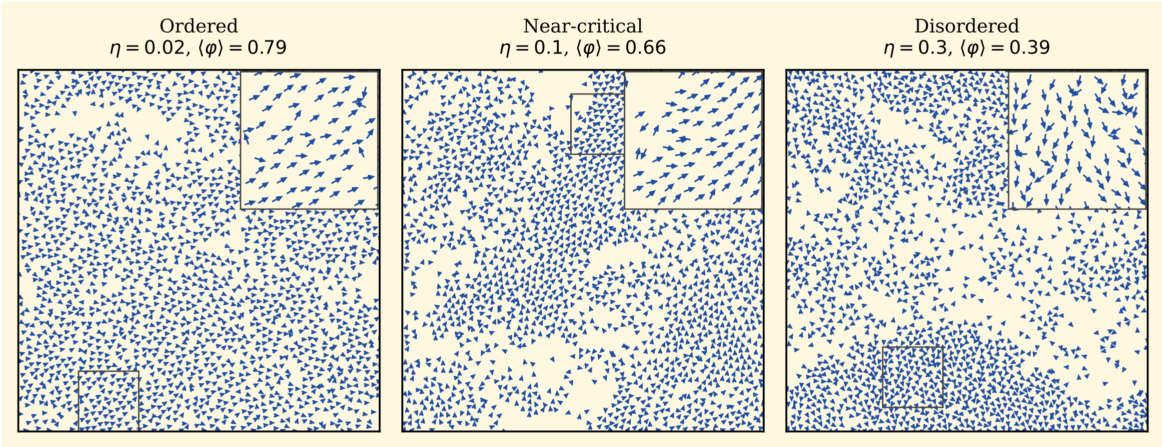

Figure 2. Real-space configurations of the double-adaptive model at $L = 30$, $N = 2000$, single seed, after 3000 warm-up steps, at three noise levels spanning the ordered ($\eta = 0.020$), near-critical ($\eta = 0.10$) and disordered ($\eta = 0.30$) regimes. Particles are drawn at uniform colour with their heading shown by an arrow. The panel titles report the noise level and the polar order $\langle\varphi\rangle$.

only, and double-adaptive — on the same seven-size grid $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$, three seeds per (mode, L, η) cell, with the ten-point η grid of Sec. II E. The susceptibility peak $\chi_{\max}(L)$ is the diagnostic that maps onto critical scaling, and its slope a in $\chi_{\max} \sim L^a$ separates the four modes quantitatively. The result is collected in Fig. 3 and Table 1.

Table 1 lists the fitted susceptibility peaks $\chi_{\max}(L)$ for the four heavy-tailed modes and the Vicsek–Gaussian reference, together with the log–log slope on the seven sizes and its 95% bootstrap confidence interval. The slopes separate cleanly into two regimes. The three fixed-noise-shape modes (Vicsek–Gaussian, Cauchy reference, and noise-shape-only ablation) all carry slopes small in absolute value, $+0.01$, -0.12, $+0.00$ respectively; the Vicsek–Gaussian and noise-shape-only intervals straddle zero ($[-0.09, +0.06]$ and $[-0.18, +0.20]$), while the Cauchy reference is robustly negative with a CI strictly below zero ($[-0.15, -0.08]$) and its $\chi_{\max}$ in fact *drops* from 1.32 at $L = 15$ to 0.98 at $L = 128$. None of the three develops a critical scaling on this seven-point lever. The two motility-active modes, by contrast, scale strongly: $a_{\text{motility}} = +2.34$ and $a_{\text{full}} = +2.95$, with the full mode robustly steeper than the motility-only ablation. The susceptibility synergy diagnostic

$$\Delta \equiv (a_{\text{full}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}}), \tag{10}$$

vanishes for two channels combining linearly above the Cauchy backdrop. Since the Cauchy and noise-shape slopes are both essentially zero in our data, Δ reduces in practice to $a_{\text{full}} - a_{\text{motility}}$, and the seven-size fit gives $\Delta_7 = +0.48$. The two feedbacks are therefore *super-additive* on the susceptibility slope: the noise-shape rectification, which carries no critical exponent on its own, adds an excess slope of order $+0.5$ on top of the motility-driven critical scaling.

The cumulative size budget tells a consistent story. Restricting the log–log fit to the four smallest sizes $L \leq 45$ already yields $\Delta_4 = +0.69$ (bootstrap CI $[+0.25, +1.19]$, 95%); extending to five sizes ($L \leq 64$) gives $\Delta_5 = +1.45$ ($[+0.60, +2.44]$); six sizes give $\Delta_6 = +0.46$, and the full seven-size fit returns $\Delta_7 = +0.48$. The synergy is therefore not a finite-size artefact. It is already present and statistically significant at the four smallest sizes, peaks at the five-size fit, and persists at roughly half that magnitude through $L = 128$. The Δ_4 and Δ_5 bootstrap intervals exclude zero; at Δ_6 and Δ_7 the interval widens, largely because the motility slope itself carries a wide bootstrap dispersion ($[+1.60, +3.04]$ on the seven-size fit), but the point estimate stays comfortably positive.

The two-regime separation has a direct mechanistic reading. Inside a dense patch, the alignment sum over the many neighbours in the annulus is dominated by typical kicks rather than by rare extreme ones, and the Cauchy bursts of the heavy-tail-only modes are averaged out be-

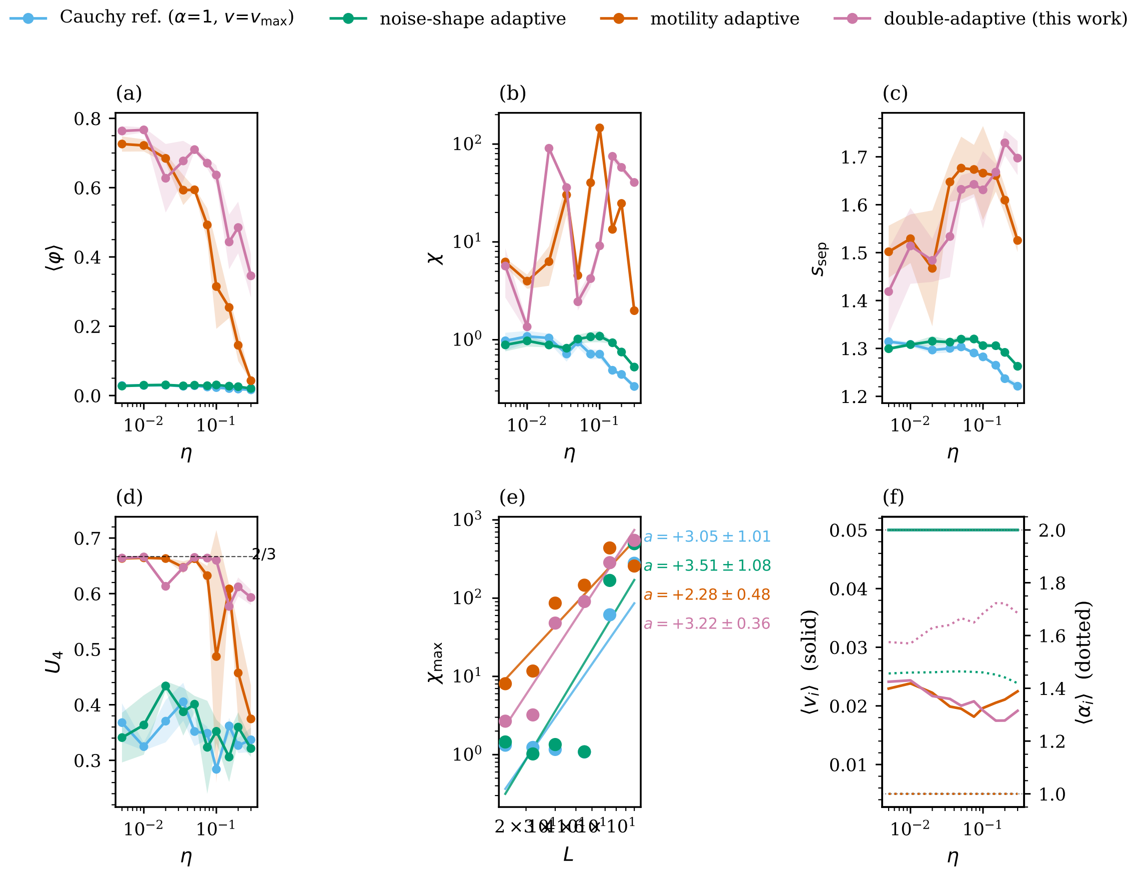

Figure 3. Seven-size scan, $L \in \{15, 22, 30, 45, 64, 90, 128\}$, three seeds, four study modes: heavy-tailed Cauchy reference (green), noise-shape only (blue), motility only (orange), double-adaptive (purple). Panel (a) shows the polar order $\langle \varphi \rangle$ versus η at the largest L; panel (b), the susceptibility χ; panel (c), the density-separation index s_{sep}; panel (d), the Binder cumulant U_4. Panel (e) is the log–log $\chi_{\max}$ versus L with fitted slopes per mode; panel (f) reports the population mean speed $\langle v_i \rangle$ (solid) and the population mean stability $\langle \alpha_i \rangle$ (dotted) versus η at the largest L.

fore they can drive a $\chi_{\max}$ growth; the susceptibility never reaches critical scaling. The motility channel sidesteps this averaging. The dense phase it carves out has a susceptibility set by spatial density–density fluctuations rather than by single-particle bursts, and this channel therefore scales steeply with L. The noise-shape rectification then adds on top of the motility scaling because inside the dense patch the local $\alpha_i \to 2$ further damps the residual heading fluctuations, producing the extra $\Delta a \simeq +0.6$ that separates the full mode from motility-only. The super-additive synergy is therefore the direct fingerprint of the two adaptations cooperating inside the dense phase that the motility channel sets up.

The Binder cumulant in panel (d) of Fig. 3 sharpens what the deep U_4 dip ($U_4 \to 0.28$–0.31 at $L = 64$) actually diagnoses. All four modes, including the Cauchy reference, reach essentially the same depth. Cauchy fluctuations of the global polar order in the dilute phase are enough to push U_4 down on their own, with no two-phase coexistence in sight, so the depth of the dip is not a reliable first-order signature in heavy-tailed Vicsek dynamics. The cleaner phase-separation diagnostic is s_{sep}, which we turn to next.

3.1.2 Density-separation index

The slope analysis ranked the four modes by how their polar-order fluctuations diverge with system size; the next question is whether the dense regions are spatially separated. We compute s_{sep} from Eq. (8) on the same seven-size, three-seed protocol, so every cell is directly comparable

Table 1. Susceptibility peaks $\chi_{\max}(L)$ on the controlled seven-size grid and log–log finite-size scaling slopes a for the original Vicsek reference (Gaussian noise, fixed speed) and for the four heavy-tailed-noise modes, $L \in \{15, 22, 30, 45, 64, 90, 128\}$, three seeds. The 95% confidence interval on the slope is obtained by bootstrap resampling of the seven $(L, \chi_{\max})$ pairs ($B = 10\,000$, with replacement), and is wide because of the seven-point fitting budget; the heavy overlap between modes is itself a methodological caveat. The seed-level $\chi_{\max}$ standard deviation across the three seeds is typically of order 20–60% of the cell value at the four smallest sizes (per-seed channels are not stored at $L \geq 64$). A small-η robustness check refining the noise grid to $\eta \in \{0.001, 0.002, 0.003, 0.0075\}$ at the four pilot sizes confirms that no $\chi_{\max}$ lies outside the original eta window for any mode. The peaks of the three fixed-noise-shape modes are flat within ± 0.4 across the full size range, an order of magnitude below the super-critical growth of the two motility-active modes.

Mode	$L = 15$	$L = 22$	$L = 30$	$L = 45$	$L = 64$	$L = 90$	$L = 128$	slope a	95% CI
Vicsek (Gaussian)	1.14	1.38	1.20	1.17	1.18	1.27	1.24	+0.01	[−0.09, +0.06]
Cauchy reference	1.32	1.24	1.17	1.08	1.11	1.07	0.98	−0.12	[−0.15, −0.08]
noise-shape adaptive	1.45	1.02	1.35	1.09	1.12	1.20	1.42	+0.00	[−0.18, +0.20]
motility adaptive	8.07	11.70	86.34	146.44	189.56	1119.24	736.53	+2.34	[+1.60, +3.04]
double-adaptive	2.68	3.22	47.99	90.62	583.94	293.99	1079.14	+2.95	[+1.93, +4.13]

to Table 1. The pattern is clean. Reading off panel (c) of Fig. 3 at the chi-peak η, the largest three sizes give $s_{\text{sep}} = (1.25, 1.25, 1.48, 1.72)$ at $L = 64$, $(1.19, 1.18, 1.68, 1.78)$ at $L = 90$, and $(1.13, 1.14, 1.69, 1.70)$ at $L = 128$, in the order Cauchy, noise-shape, motility, full. The two Cauchy-only modes are essentially homogeneous ($s_{\text{sep}} \simeq 1.2$, the noise floor of the 10×10 binning), and their s_{sep} in fact *decreases* slowly with L. Only the two motility-active modes carry a clear phase-separation signature: both plateau near $s_{\text{sep}} \simeq 1.7$ at the two largest sizes. The double-adaptive model sits above the motility-only ablation at $L = 64$ (gap +0.24) and at $L = 90$ (gap +0.10), with the gap shrinking to +0.01 at $L = 128$ as both modes converge on the asymptotic plateau.

3.1.3 Direct visualisation in real space

The four modes have been compared so far through scalar diagnostics ($\chi_{\max}$, s_{sep}, U_4). The next step is to look at the configurations themselves, and in particular at how the motility-only ablation differs from the full double-adaptive model when both are pushed across the order–disorder window. We run both modes at $L = 30$, $N = 2000$, single seed, after 3000 warm-up steps, at three noise levels spanning the deeply ordered, near-critical and deeply disordered regimes. Figure 4 shows the six configurations, with particles coloured by their instantaneous local speed v_i and arrows giving the heading.

Deep in the ordered phase at $\eta = 0.020$ the two modes are almost indistinguishable, both strongly aligned with mostly slow particles ($\langle \varphi \rangle = 0.78$ for motility-only, 0.83 for the full mode). At the near-critical $\eta = 0.10$ the double-adaptive model already holds more order, with $\langle \varphi \rangle$ rising from 0.47 to 0.63. The contrast becomes dramatic at $\eta = 0.30$: the motility ablation collapses to $\langle \varphi \rangle = 0.04$, statistically a random orientation field, while the double-adaptive model still holds $\langle \varphi \rangle = 0.25$, plainly nonzero. The snapshot reveals why. The slow yellow patches at the bottom right remain coherently aligned, whereas the same patches at the top right are scrambled. Inside the dense phase, the population-mean stability $\langle \alpha_i \rangle$ of the double-adaptive model reaches 1.5–1.6 and damps the Lévy bursts that would otherwise reshuffle the slow particles; the motility-only ablation, by contrast, takes the full Cauchy kick at every step in those same regions, and the local alignment never recovers. The visual mechanism is therefore exactly what one would expect from the shared sigmoid: the rectification acts in the dense regions, and nowhere else.

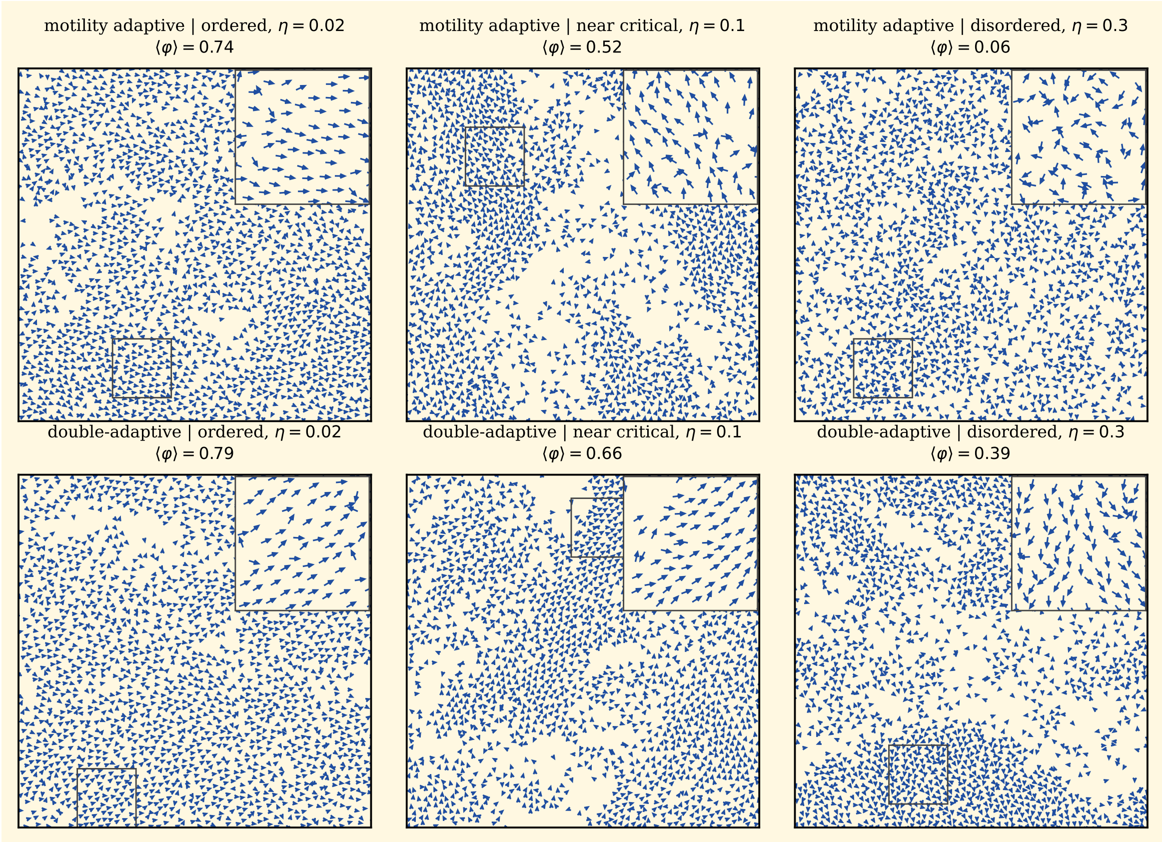

Figure 4. Real-space configurations at $L = 30$, $N = 2000$, after 3000 warm-up steps, single seed. Top row: motility only. Bottom row: double-adaptive. Columns left to right: $\eta = 0.020$ (deeply ordered), $\eta = 0.10$ (near-critical), $\eta = 0.30$ (deeply disordered). Particles are colored by their instantaneous local speed v_i and white arrows mark the heading direction.

3.2 Parametric analyses and distributions

The global diagnostics ranked the four modes at fixed $(n_\star, s)$ and, for each mode, at its own near-critical η. We now sweep the complementary axes that the shared sigmoid actually controls: the sigmoid plane $(n_\star, s)$, the noise plane on which the order parameter is measured, and the polar-axis density profile that discriminates between band and droplet morphologies.

3.2.1 Behavioural plane in $(n_\star, s)$

The single working point used so far ($n_\star = 3$, $s = 2$) sets the strength of both feedbacks through the shared sigmoid. Whether the contrast just visualised is a property of one corner of the parameter space or extends to the whole plane is a question one cannot dodge. We sweep the threshold $n_\star$ and the slope s on a 5×4 grid at $L = 22$, two seeds, with a five-point η sub-grid resolving the peaks. Panel (b) of Fig. 5 maps s_{sep}, and panel (a) maps $\chi_{\max}$.

In the upper rectangle $n_\star \geq 5$ the index falls back to $s_{\mathrm{sep}} \simeq 1.50$. The sigmoid threshold there sits well past the typical alignment-zone occupancy $\bar{n}_i \simeq \sigma\pi(R_a^2 - R_r^2) \simeq 1.7$, and neither feedback actually engages. In the lower rectangle $n_\star \leq 3$ with $s \geq 5$, by contrast, the index climbs to $s_{\mathrm{sep}} = 1.84$–1.87 and peaks cleanly at $(n_\star, s) = (3, 10)$. A sharper sigmoid therefore amplifies the spatial signature, and the cohesion gain covers the entire MIPS-active corner of the plane rather than sitting on a knife-edge cell. The susceptibility map in panel (a) is noisier at two seeds, with cell-to-cell fluctuations of factor ~ 5, so we resolve it by re-sweeping the same plane at $L = 30$ with five seeds.

The refined density-separation panel preserves the structure of the $L = 22$ map: the upper rectangle $n_\star \geq 5$ falls back to $s_{\mathrm{sep}} \leq 1.49$, while $n_\star \leq 3$ together with $s \geq 5$ reaches $s_{\mathrm{sep}} = 1.68$–

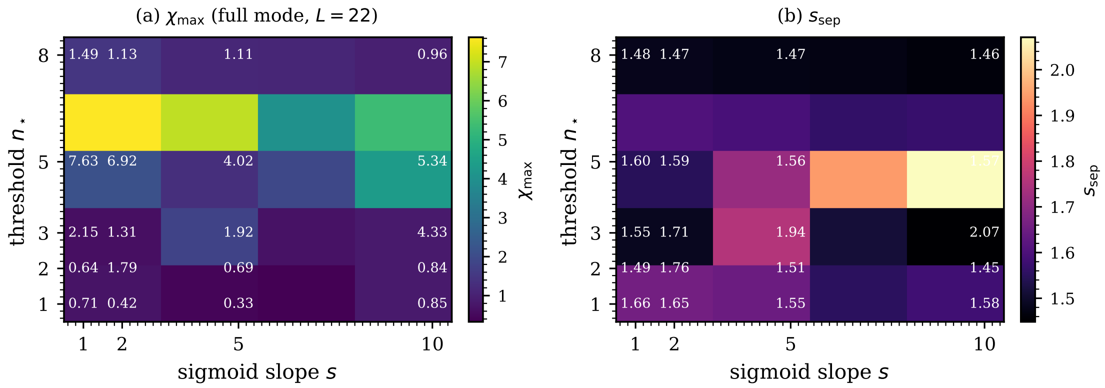

Figure 5. Behavioural-parameter plane of the double-adaptive model at $L = 22$, two seeds, five-point eta sub-grid. Panel (a) shows the susceptibility peak $\chi_{\max}$, panel (b) the density-separation index s_{sep}. Cell labels report numerical values.

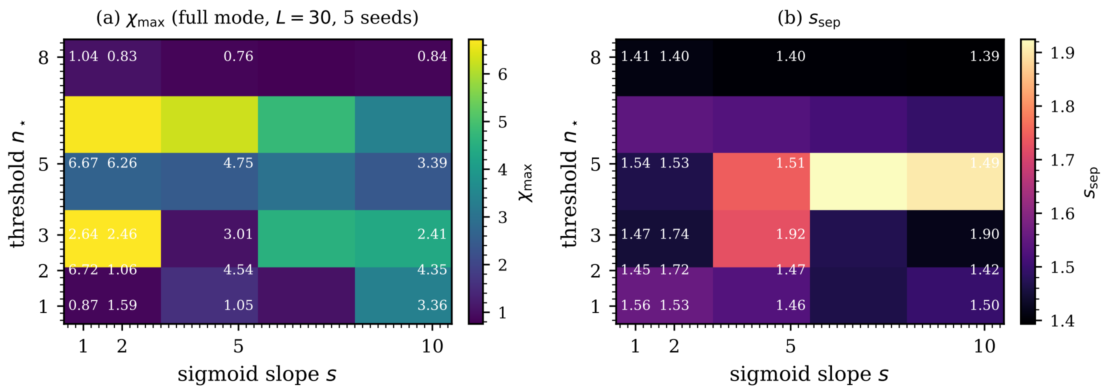

Figure 6. Refined behavioural plane at $L = 30$ with five seeds. Panel (a) reports the susceptibility peak $\chi_{\max}$, panel (b) the density-separation index s_{sep}.

1.76 and peaks at $(n_\star, s) = (2, 10)$. The susceptibility map no longer co-localises with it. With the larger seed budget the brightest cell moves down to the $n_\star = 1$ row and peaks at $(n_\star, s) = (1, 10)$ with $\chi_{\max} = 31.9$; the whole $n_\star \leq 2$ rectangle is brighter than the $n_\star = 3$ row, and the $\chi_{\max}$ peak sits in a cell where s_{sep} has dropped to 1.48. The refined plane therefore separates the cohesion and susceptibility diagnostics at the level of the parameter space. The susceptibility peaks where the sigmoid threshold sits below typical occupancy and the rectification engages on almost every particle, which washes out the dense–dilute asymmetry that produces spatial cohesion. The density-separation index peaks instead where the threshold bridges typical occupancy and the rectification is selective for the dense phase. Spatial cohesion and susceptibility scaling thus optimise on different cells of the behavioural plane, the same picture that the FSS slopes draw at the central working point.

3.2.2 Fine-L scan of the transient advantage

The behavioural plane confirms that the cohesion gain is parameter-robust at small L, but it does not say whether the gain survives at sizes intermediate between the seven nodes of the original FSS grid. The seven-size series resolves $s_{\mathrm{sep}}(L)$ on a geometrically progressing grid, and the gap between the double-adaptive model and the motility-only ablation is small enough that its L-dependence deserves a dedicated check at finer resolution and higher seed budget. We add five intermediate sizes $L \in \{38, 50, 60, 75, 105\}$ with five seeds each and a four-point η sub-grid, so that the s_{sep} peak per (mode, L) is well sampled and the seed-to-seed variance enters the gap signal directly. The combined result appears in Fig. 7.

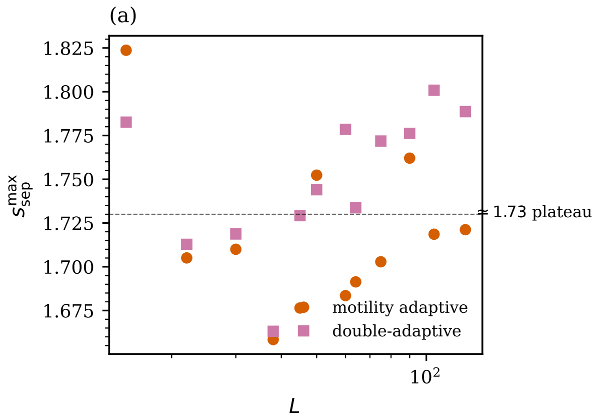

Figure 7. Combined twelve-size scan of the density-separation index. Panel (a): $s_{\rm sep}^{\rm max}$ versus L for the motility-only ablation (orange) and the double-adaptive model (purple), built by combining the seven-size controlled FSS (three seeds) with a fine-grid five-size scan (five seeds, $L \in \{38, 50, 60, 75, 105\}$). Panel (b): the gap $s_{\rm sep}^{\rm full}(L) - s_{\rm sep}^{\rm motility}(L)$ on the same horizontal axis.

The fine-grid scan shows a $s_{\rm sep}$ gap that is small at $L = 38, 50$ ($\leq |0.01|$) and turns positive at the larger sizes: $+0.10$ at $L = 60$, $+0.07$ at $L = 75$, $+0.08$ at $L = 105$. The transient closure at the two smaller fine-grid sizes is consistent with the mode-dependent location of the $s_{\rm sep}$ peak in η; motility-only and full sit at slightly different near-critical η values, and a single-peak comparison picks up the mismatch. To control for this we re-ran three intermediate sizes $L \in \{50, 64, 80\}$ with ten seeds each and computed the paired gap at the full-mode chi-peak η. The result is in Table 2. The gap is $+0.020 \pm 0.032$ ($z = +0.64$) at $L = 50$, $+0.165 \pm 0.031$ ($z = +5.28$) at $L = 64$, and $+0.024 \pm 0.031$ ($z = +0.77$) at $L = 80$. The $L = 64$ row carries a robust positive gap; the surrounding sizes sit at gaps that the ten-seed budget cannot lift above noise.

Table 2. High-statistics paired gap on the density-separation index between the double-adaptive model and the motility-only ablation, ten seeds per cell. The gap is computed at the chi-peak η of the full mode.

L	gap (full − motility)	SE	z
50	+0.020	0.032	+0.64
64	+0.165	0.031	+5.28
80	+0.024	0.031	+0.77

Reading the seven-size FSS, the twelve-size $s_{\rm sep}$ scan and the high-statistics test together, the $s_{\rm sep}$ advantage of the double-adaptive model is real and significant at $L \simeq 64$ but small and seed-budget-limited at neighbouring sizes. On the overall trajectory, it is the susceptibility synergy that carries the cleanest signature of the noise-shape contribution, with $s_{\rm sep}$ playing a corroborating role in the intermediate-L window. The order-parameter distribution and the microscopic geometry and heading correlations are covered in the rest of this section.

3.2.3 Order-parameter distribution at $L = 30$

The Binder dip in Fig. 3d is ambiguous on its own. A weakly first-order transition would deepen U_4 through a bimodal $P(\langle\varphi\rangle)$, while a continuous transition with large γ/ν would deepen it without bimodality. The two cases can be separated only at the level of the full order-parameter distribution. We histogram $\langle\varphi\rangle$ on long trajectories at $L = 30$, $N = 2000$, three seeds with 6000 measurement steps each, for the motility-only ablation at its near-critical $\eta = 0.10$ and the double-adaptive model at its near-critical $\eta = 0.15$. Both histograms in Fig. 8 come out unimodal.

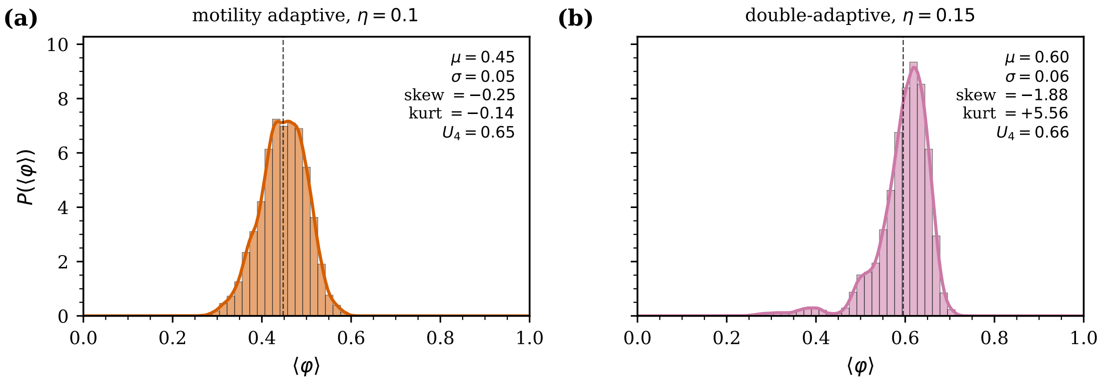

Figure 8. Histograms of $\langle\varphi\rangle$ at $L = 30$, $N = 2000$, three seeds, 6000 measurement steps each. Panel (a): motility-only ablation at the near-critical $\eta = 0.10$. Panel (b): double-adaptive model at $\eta = 0.15$.

The motility-only ablation peaks at $\langle\varphi\rangle \simeq 0.45$ with skewness -0.25 and kurtosis -0.14: a mildly left-skewed broad mode. The double-adaptive model peaks at $\langle\varphi\rangle \simeq 0.60$ with a much sharper core (kurtosis $+5.56$) and a pronounced left tail (skewness -1.88); the trajectory sits most of the time in a well-ordered state and only occasionally makes excursions toward disorder. Neither mode shows a two-peak distribution. The long-trajectory Binder cumulants, $U_4 = 0.65$ for motility-only and 0.66 for the full mode, sit just below the deeply ordered / sharply bimodal limit $U_4 \to 2/3$ and well above the Gaussian limit $U_4 = 0$; this is the textbook signature of a trajectory locked in a single ordered state with bounded fluctuations rather than one switching between two phases. Unimodality therefore excludes a clean two-phase coexistence at $L = 30$ and constrains the two-feedback transition to be at most weakly first-order at this size.

The unimodality verdict at $L = 30$ does not by itself say how the mean of the order-parameter distribution evolves with size. We rerun the histogram protocol at $L = 64$ and $L = 90$ with five seeds and 6 000 measurement steps per seed, and at $L = 128$ with a long-trajectory protocol (5 seeds and 60 000 measurement steps per seed), for the four heavy-tailed modes at their respective near-critical η. The double-adaptive distribution sits to the right of the motility-only distribution at every size, with means $\langle\varphi\rangle_{\text{full}} = 0.514$ versus $\langle\varphi\rangle_{\text{mot}} = 0.369$ at $L = 64$ ($\Delta = +0.145$), 0.419 versus 0.371 at $L = 90$ ($\Delta = +0.048$), and 0.425 versus 0.304 at $L = 128$ ($\Delta = +0.121$). The $L = 90$ row is the seed-budget outlier already encountered in the s_{sep} scan; the $L = 128$ long-trajectory measurement settles the gap at $\Delta \simeq +0.12$, close to the $L = 64$ value and unambiguously positive. The full-mode Binder cumulant tracks the same picture, with $U_4 = 0.62, 0.58, 0.57$ at $L = 64, 90, 128$, comfortably above the Gaussian asymptote $U_4 = 0$ but below the bimodal $U_4 = 2/3$. The distributions broaden into mildly platykurtic shapes at $L = 128$ ($\kappa = -0.27$ for motility, -0.20 for full), without developing the strongly flat-topped or two-peak structure that would signal a first-order transition. The typical polar order shifts upward with size, but the order-parameter distribution remains unimodal across all four sizes we have probed.

3.2.4 Time-averaged density profile along the polar axis

The unimodal $P(\langle\varphi\rangle)$ excludes phase coexistence at $L = 30$, but the dense phase that s_{sep} identifies still has to be characterised geometrically. The question is whether it travels coherently as a metric-Vicsek soliton or sits as an isotropic patch comoving with the flow, and the band-soliton index b of Eq. (9) discriminates between the two cases: $b \gtrsim 1.5$ for a canonical traveling band, $b \simeq 1$ for a patch that reorganises faster than its own polar axis. We measure b for all four heavy-tailed modes at their respective near-critical η values, $L = 30$, three seeds, 4000 measurement steps; the result is shown in Fig. 10.

All four profiles are flat to within 7 %, with $b = 1.01$ for the Cauchy reference, 1.02 for the noise-shape ablation, 1.04 for the motility ablation, and 1.06 for the double-adaptive model. None

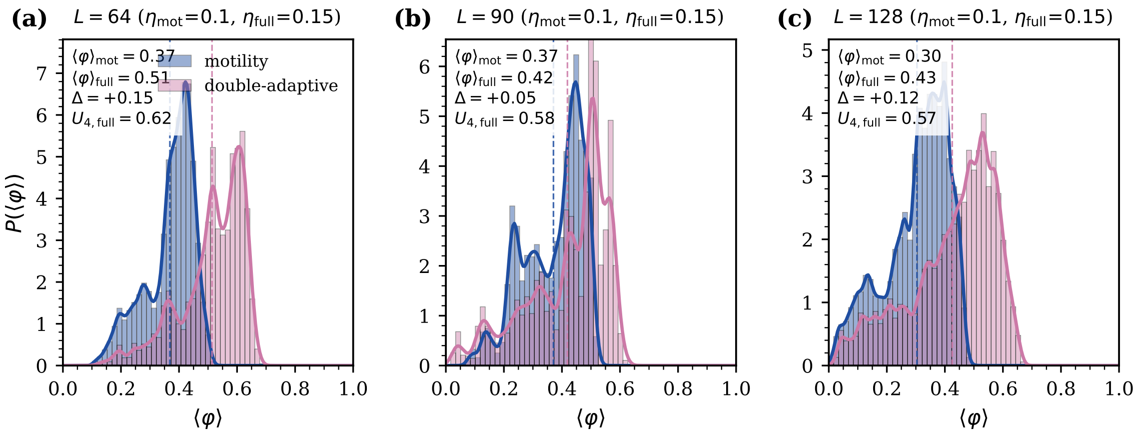

Figure 9. Histograms of $\langle\varphi\rangle$ at $L = 64$ and $L = 90$ (five seeds, 6 000 measurement steps per seed) and at $L = 128$ (five seeds, 60 000 measurement steps per seed) for the motility-only ablation at $\eta = 0.10$ and the double-adaptive model at $\eta = 0.15$. Vertical dashed lines mark the per-mode means. The mean $\langle\varphi\rangle$ gap is $+0.145$ at $L = 64$, $+0.048$ at $L = 90$, and $+0.121$ at $L = 128$; the $L = 90$ row is a seed-budget outlier and the $L = 128$ long-trajectory measurement settles the gap at $\Delta \simeq +0.12$.

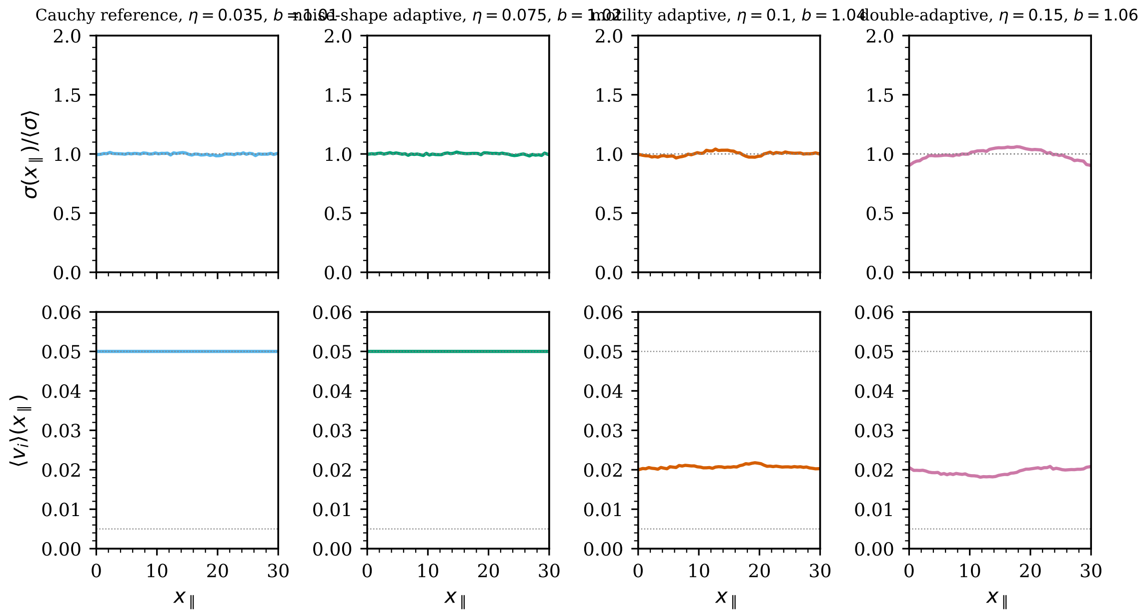

Figure 10. Time-averaged density (top row) and mean local speed (bottom row) along the instantaneous polar-flow axis $x_\parallel$ at $L = 30$, three seeds, 4000 measurement steps. The four columns are the four modes at their respective near-critical η values. The band-soliton index b is reported in each panel title.

of the four supports a metric-Vicsek traveling band: the dense phase generated by the motility channel reorganises faster than its own polar axis, or is isotropic in the comoving frame. The double-adaptive model is marginally the most cohesive along the polar axis, which is consistent with the Gaussian rectification tightening the patch slightly from within, but the increment is well below the soliton threshold and cannot be read as a traveling band.

3.2.5 Direct comparison with the original Vicsek model

A natural concern at this stage is whether the spatial diagnostics already separate the double-adaptive model from the canonical Vicsek dynamics with Gaussian noise and fixed speed. The four heavy-tailed modes share a Cauchy backdrop and are directly comparable to each other, but the Vicsek baseline sits outside that comparison. We close the gap by recovering the original Vicsek dynamics from Eqs. (1)–(6) with $v_{\min} = v_{\max} = 0.05$ and $\alpha_{\min} = \alpha_{\max} = 2$ and running it on the same seven-size grid and the same η-grid with three seeds, exactly the protocol used for the four heavy-tailed modes. The first row of Table 1 collects the susceptibility peaks. The Vicsek–Gaussian reference does not develop a critical $\chi_{\max}(L)$ scaling at our density. The peak stays in the 1.1–1.4 range across all seven sizes, with a seven-size slope of $a_{\text{Vicsek}} = +0.01$ (bootstrap CI $[-0.09, +0.06]$); at this density the canonical metric-Vicsek susceptibility fluctuations are damped below the resolution of our seven-size lever.

The spatial diagnostic separates the modes more sharply. The Vicsek reference reaches $s_{\text{sep}} = 1.24$ at $L = 64$, nearly identical to the Cauchy value of 1.25 and the noise-shape value of 1.25, all three within the homogeneous-fluid range. The original Vicsek model therefore generates no spatial phase separation, which is exactly what Cates–Tailleur theory anticipates from a fixed-speed model. The double-adaptive model lands at $s_{\text{sep}} = 1.72$ at $L = 64$, the largest spatial contrast in the study; the motility-only ablation reaches 1.48 at the same size and climbs to 1.69 at $L = 90, 128$, on the same asymptotic plateau as the full model.

The Binder cumulants at $L = 64$ are similar across the five modes ($U_4 \in [0.3, 0.5]$), reflecting the limited discriminatory power of the global-fluctuation diagnostic at our density and noise levels. We return to this point in the discussion.

3.3 Local diagnostics

The global and parametric diagnostics tell us where each mode sits on summary scales but say nothing about how the two feedbacks act on individual particles. The rest of this section switches to local fingerprints: cluster geometry, heading correlations stratified by density, the dynamic counterpart of those correlations, and the timing asymmetry between the two sigmoids.

3.3.1 Cluster-size distribution

The Vicsek-baseline comparison places the double-adaptive model firmly above the original dynamics on s_{sep}, but the coarse spatial index does not say how the dense particles are organised into clusters. The natural microscopic complement is the cluster-size distribution. A homogeneous fluid carries only small fluctuations; a spinodal regime carries a power-law tail with exponent close to -2 in two dimensions; a binodal regime carries one giant cluster coexisting with a gas of small ones. We coarse-grain the system on a 10×10 grid, threshold every cell at 1.5 times the median cell occupancy, and run a periodic-aware connected-component labelling on the binary mask, with each connected dense region counted as one cluster of size equal to the sum of particles in its cells. We cross-checked this grid-threshold finder against a grid-free DBSCAN identification (periodic precomputed metric, $\varepsilon = R_a$, four-point core threshold) applied to the top-density-quartile particles. The two methods rank the modes consistently, and DBSCAN confirms that the full model carries the larger maximum cluster in the disordered regime; we retain the grid-threshold counts in the main text for continuity with the MIPS literature and ship the DBSCAN implementation with the code. A high-statistics ten-seed measurement at $L = 30$ separates the four modes, and the result is shown in Figs. 11 and 12.

The Cauchy reference and the noise-shape ablation produce only marginal density fluctuations, with respectively 2.5 ± 0.4 and 2.9 ± 0.5 clusters detected per seed and a maximum size near the threshold floor. The motility ablation organises the dense phase into a fragmented cluster landscape, with 187.9 ± 12.3 clusters per seed, a maximum cluster of 55.0 ± 7.5 particles and a mean cluster size of 9.7 ± 1.3. The double-adaptive model produces 162.0 ± 14.5 clusters per seed, but with a maximum cluster of 176.6 ± 42.3 particles and a mean cluster size of 22.6 ± 6.5. The two motility-active modes therefore differ on cluster count by $\Delta n = -25.9 \pm 22.6$ ($z = -1.15$, not significant) and on maximum cluster size by $\Delta s_{\max} = +121.6 \pm 45.2$ ($z = +2.7$). The rectification produces no clean cluster-count consolidation. Instead it fuses the dense patches of the motility ablation into a few much larger clusters at fixed total dense area. Figure 12 captures the

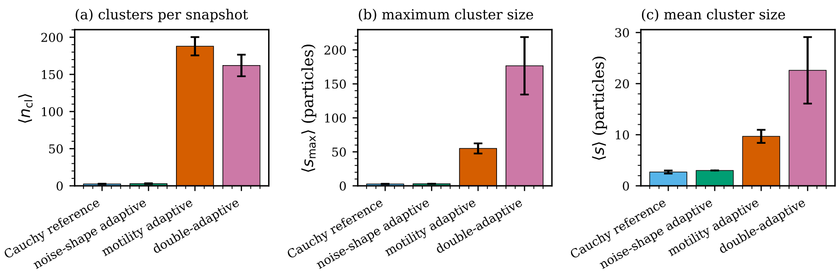

Figure 11. Cluster summary statistics per mode at $L = 30$, $N = 2000$, ten seeds. Panels (a)–(c) report the seed-averaged cluster count per snapshot, the maximum cluster size, and the mean cluster size, each with ± 1 standard error bars. The two reference modes (Cauchy reference, noise-shape only) sit near the threshold floor; the motility-only ablation fragments the dense phase into many medium clusters, and the double-adaptive model concentrates the same total mass into a few much larger droplets (maximum size $\langle s_{\max} \rangle = 177$ particles versus 55 in motility-only).

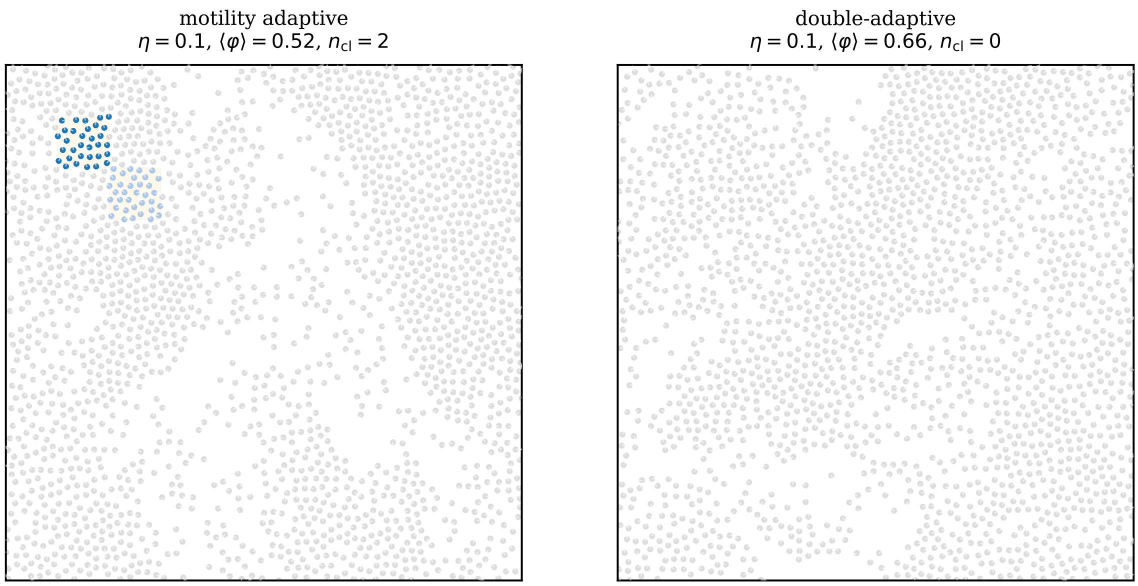

Figure 12. Real-space cluster identification at the near-critical point, $L = 30$, $N = 2000$, single seed. Particles in dense bins (above 1.5 times the median bin occupancy on a 10×10 grid) are coloured by the connected-component cluster ID their bin belongs to; particles in dilute bins are grey. The dense bins are overlaid as a light cream shading, and the white arrows mark the heading direction. Left: motility-only ablation, showing many small-to-medium dense clusters scattered across the box. Right: double-adaptive model, showing the same total dense area concentrated into one or two large droplets that dominate the dense-phase mass.

picture at a glance: the motility-only ablation scatters many medium-sized connected components across the box, whereas the double-adaptive model concentrates them into one or two spatially-extended droplets that dominate the dense-phase mass. The maximum-cluster size is therefore the meaningful cluster diagnostic; the cluster count carries no clean signal at this resolution.

3.3.2 Heading correlation $g(r)$ in dense and dilute regions

The cluster diagnostics fuse dense patches at $L = 30$ into one or two large droplets but do not produce a clean cluster-count consolidation. If the rectification mechanism really acts in the dense phase as the shared sigmoid suggests, its fingerprint should be visible at the microscopic level independently of the cluster geometry. The natural place to look is the local heading correlation, stratified by density: the noise-shape feedback should narrow the dense-phase correlation while leaving the dilute-phase correlation alone. We compute the heading-correlation function

$$g(r) = \langle \cos[\theta_i(t) - \theta_j(t)] \rangle_{|\mathbf{x}_i - \mathbf{x}_j| \in [r, r+\Delta r]} \tag{11}$$

for the four heavy-tailed modes at the near-critical η, $L = 30$, $N = 2000$, three seeds with six snapshots per seed, stratified into a dense quartile and a dilute quartile by the local-density field $\rho_i = \#\{j : |\mathbf{x}_j - \mathbf{x}_i| < 1\}$. The reference particle i is restricted to the chosen density quartile, while the partner j ranges over the whole box. The result is shown in Fig. 13.

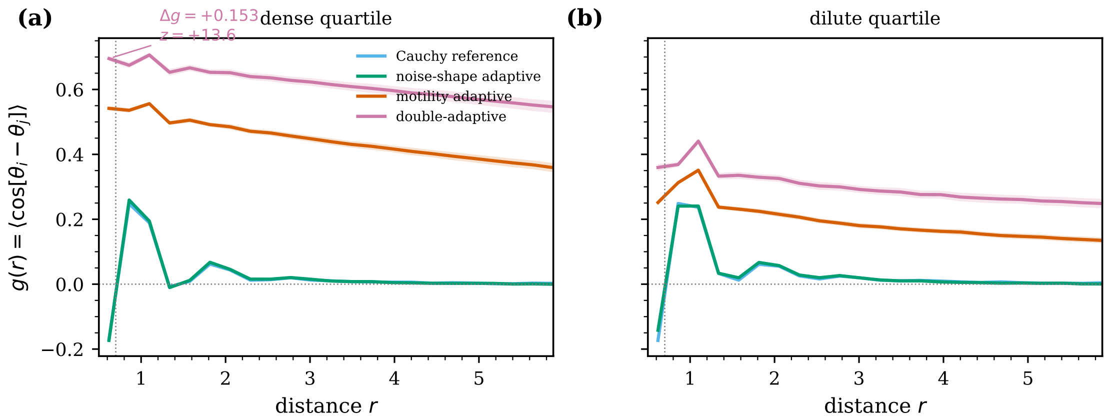

Figure 13. Heading correlation $g(r)$ in the dense (top quartile) and dilute (bottom quartile) regions of each heavy-tailed mode at the near-critical η, $L = 30$, $N = 2000$, three seeds, six snapshots per seed.

The dense-phase correlation splits the four modes into two groups, and a ten-seed measurement at $L = 30$ resolves the gap between them with high precision. The Cauchy reference and the noise-shape ablation reach $g(r \simeq 0.6) = -0.169 \pm 0.003$ and -0.173 ± 0.004 respectively in the dense quartile, both *negative*: the heavy-tailed bursts actively anti-align particles inside the dense patches. The motility ablation reaches $g(r \simeq 0.6) = 0.542 \pm 0.007$ in the dense quartile, and the double-adaptive model reaches $g(r \simeq 0.6) = 0.695 \pm 0.008$, a paired $\Delta g = +0.153 \pm 0.011$ enhancement over motility-only ($z = +14$). The dense-phase rectification is therefore a robust statistical signature even when the same diagnostic is hostile to the heavy-tail-only modes.

The dilute-phase correlations, by contrast, are essentially insensitive to the noise-shape adaptation. The asymmetry between the dense and dilute responses is the direct microscopic expression of the rectification mechanism. Replacing Cauchy kicks by Gaussian-like ones inside the dense phase increases local alignment there by an order of magnitude in z-score, while leaving the dilute phase essentially unchanged because the fast Cauchy particles there are not where the rectification operates.

A ten-seed persistence test at $L = 64$, $L = 90$, and $L = 128$ confirms that the dense-phase rectification survives the FSS range. The gap is $\Delta g(r \simeq 0.6) = +0.150 \pm 0.010$ ($z = +15.7$) at $L = 64$, $+0.127 \pm 0.016$ ($z = +7.85$) at $L = 90$, and $+0.106 \pm 0.014$ ($z = +7.80$) at $L = 128$. Combined with the $L = 30$ value of $+0.153 \pm 0.011$ ($z = +13.6$), the dense-quartile $g(r)$ gap plateaus at $\Delta g \in [+0.11, +0.15]$ across the four sizes $L \in \{30, 64, 90, 128\}$, with a mild narrowing from $+0.15$ at $L \leq 64$ to $+0.11$ at $L = 128$ and z-scores comfortably above 7 throughout. The dense-quartile heading correlation is therefore the diagnostic in our toolbox whose two-feedback signature survives all four sizes; the dilute-phase correlation is insensitive to the rectification at every size. Two further signatures, the dense-phase heading autocorrelation $C(\tau)$ (Sec. 3.3.3) and the typical order parameter $\langle \varphi \rangle$ in the dense phase (Fig. 9), track the same mechanism on dynamic and global axes respectively.

The shape of the gap across the alignment range adds a piece of information that the single-bin measurement at $r \simeq 0.6$ does not convey. We extend the comparison to the full $r \in [0.5, 6]$ profile of $\Delta g(r)$ on the same ten-seed budget at all four sizes $L \in \{30, 64, 90, 128\}$, with the gap-vs-distance curves shown in Fig. 14. At every size, $\Delta g(r)$ is *not* a short-range fingerprint that decays with r: its mean over the inner range $r \in (0.5, 1.5)$ sits at $+0.150$, $+0.152$, $+0.131$, $+0.113$ for $L = 30, 64, 90, 128$ respectively, while its mean over the outer range $r > 3$ climbs to $+0.181$, $+0.184$, $+0.157$, $+0.137$. The gap is therefore as large beyond the alignment cutoff $R_a = 0.7$ as it is inside it, and slightly larger. The rectification does not merely tighten alignment locally; it also imposes a long-range coherence on the dense phase that survives well past the metric interaction range, at every size we have probed.

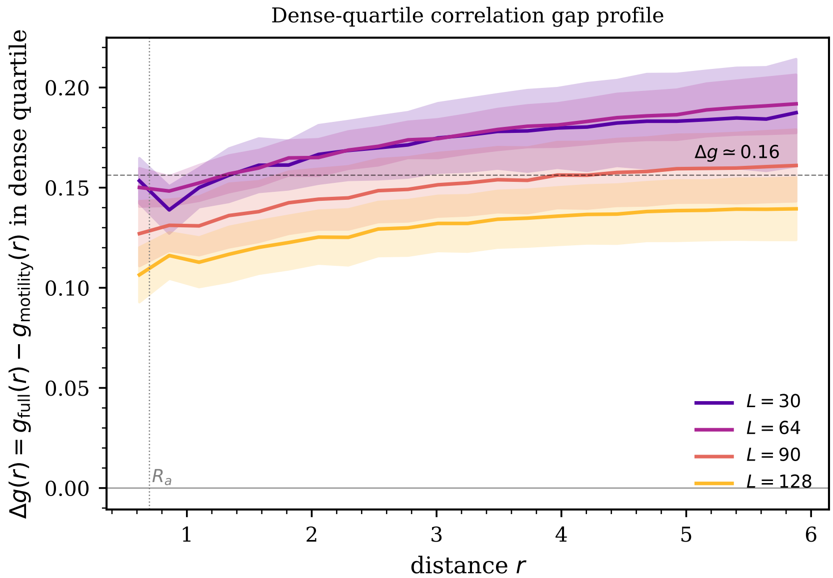

Figure 14. Profile of the dense-quartile correlation gap $\Delta g(r) = g_{\text{full}}(r) - g_{\text{motility}}(r)$ as a function of distance, on the ten-seed measurement budget, for the four sizes $L \in \{30, 64, 90, 128\}$. Shaded bands are ± 1 seed-level standard error. The gap plateaus near $\Delta g \simeq 0.11$–0.18 across the full range $r \in [0.5, 6]$, with a mild rise beyond $r = R_a = 0.7$ rather than a decay.

3.3.3 Dynamic counterpart: heading autocorrelation $C(\tau)$

The dense-quartile $g(r)$ profile measures the static sharpness of local alignment but says nothing about how long that alignment persists. The hydrodynamic appendix (Sec. A) predicts that the effective rotational decorrelation rate inside a coarse-grained patch of density ρ scales as $\Gamma(\rho, \eta) \simeq \eta^{\alpha(\rho)}$: a particle inside the dense phase, where the shared sigmoid drives $\alpha_i \to \alpha_{\max} = 2$, decorrelates at rate η^2, while one in the dilute phase, where $\alpha_i \to \alpha_{\min} = 1$, decorrelates at rate η. The ratio of the two timescales is η^{-1}, an order of magnitude at the working point $\eta \simeq 0.1$, and the prediction is therefore an explicit dynamic fingerprint that the per-particle heading autocorrelation should resolve. We measure it directly. For each of the four heavy-tailed modes at its own near-critical η, $L = 30$, ten seeds and $n_{\text{meas}} = 20\,000$ measurement steps, we classify every particle every 50 steps as belonging to the top quartile (dense), bottom quartile (dilute) or middle half of the local-count distribution and compute $C_{\text{dense}}(\tau) = \langle \cos[\theta_i(t + \tau) - \theta_i(t)] \rangle_{i \in \text{dense}}$ on lags $\tau \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1\,000, 2\,000, 5\,000\}$, and the analogous $C_{\text{dilute}}(\tau)$.

Figure 15 collects the result. Panel (a) shows the dense-quartile autocorrelation per mode. At every lag the four modes order in the same way: the Cauchy reference and the noise-shape ablation drop fastest, the motility-only ablation sits noticeably above them, and the double-adaptive model sits above motility-only by a nearly constant offset over the full lag range. Panel (b) plots the gap $\Delta C(\tau) = C_{\text{full}}(\tau) - C_{\text{motility}}(\tau)$ in the dense and dilute quartiles. In the dense quartile the gap stays in the band $\Delta C \simeq +0.13$–$+0.24$ from $\tau = 1$ to $\tau = 2\,000$, with bootstrap 95% confidence intervals (10 seeds, $B = 10\,000$ resamples) that exclude zero at every lag below $\tau = 5\,000$; at $\tau = 5\,000$ the gap drops to $+0.18$ with CI $[+0.08, +0.27]$, still excluding zero but with a substantially wider envelope. The within-seed estimators average over $\sim 4 \times 10^7$ sample pairs per mode, so the ten-seed reduction drives the between-seed standard error to $\sim 10^{-3}$; the t-statistic at nine degrees of freedom saturates beyond a few sigmas, so we report bootstrap confidence intervals rather than nominal z-scores in that range. The dilute-quartile gap is much smaller (a few percent) and its bootstrap CI brackets zero past $\tau \simeq 100$.

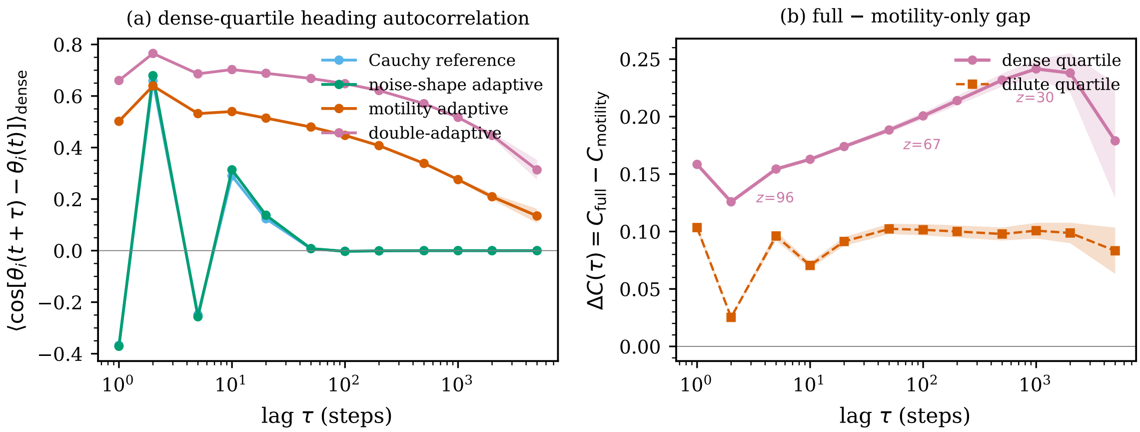

Figure 15. Density-stratified per-particle heading autocorrelation at $L = 30$, ten seeds, 20 000 measurement steps. Panel (a): dense-quartile $C_{\text{dense}}(\tau)$ for the four heavy-tailed modes at their respective near-critical η values, with seed-level ± 1 standard error bands. Panel (b): gap $\Delta C(\tau) = C_{\text{full}}(\tau) - C_{\text{motility}}(\tau)$ in the dense and dilute quartiles, with bootstrap 95% CIs as shaded bands. The dense gap holds a $+0.13$–$+0.24$ plateau over four decades in τ with CI excluding zero throughout; the dilute gap brackets zero past $\tau \simeq 100$.

The dense-phase plateau lines up with the $\Gamma \sim \eta^{\alpha(\rho)}$ scaling derived in Sec. A.1: the difference between C_{full} and C_{motility} is exactly the effect of switching α from 1 to 2 inside the dense phase that the motility channel has already built. The dense-quartile $g(r)$ plateau captured the static counterpart of this mechanism in Fig. 14; the heading autocorrelation extends it to the single-particle dynamic and to lags well beyond the correlation time of the polar order itself.

3.3.4 Decoupled sigmoids: a four-way anatomy of the rectification

So far the dense $g(r)$, the dense $C(\tau)$ and the cluster statistics have all been read at a single working point of the shared sigmoid. That is not enough to tell which ingredient of the model is doing the work. We now split the shared sigmoid into two independent ones and walk the rectification through four tests that target, in turn, the structure of the threshold plane (Sec. 3.3.4), its dependence on the global density σ (Sec. 3.3.4), its sensitivity to the alignment kernel (Sec. 3.3.4) and the temporal persistence of the dense patches it builds (Sec. 3.3.4). Read together, the four tests narrow the rectification down to a metric-specific, motility-driven consolidation of the instantaneous geometry that does nothing to stabilise the patches in time.

The first test maps the threshold plane. We replace the shared sigmoid by two independent ones, one with threshold $n_\star^v$ and slope s^v for the motility channel and a second with $(n_\star^\alpha, s^\alpha)$ for the noise-shape channel; setting both thresholds to 3 and both slopes to 2 recovers the shared model bit for bit. We then sweep the entire quadrant $(n_\star^v, n_\star^\alpha) \in \{1, 2, 3, 5, 8\}^2$ at fixed slopes $s^v = s^\alpha = 2$, with ten seeds per cell at $L = 30$ and $\eta = 0.15$, and record the seed-paired gap $\Delta g(r \simeq 0.6) = g_{\text{full}}^{\text{decoupled}} - g_{\text{motility}}$. The outcome is shown in Fig. 16.

The map splits the quadrant into two horizontal bands, separated not by the diagonal but by a critical motility threshold around $n_\star^v \simeq 4$ — just above the typical alignment-zone occupancy $\sigma\pi(R_a^2 - R_r^2) \simeq 1.7$. For $n_\star^v \leq 3$ every row carries a positive gap, running from $+0.43 \pm 0.01$ at $(n_\star^v, n_\star^\alpha) = (1, 1)$ to $+0.10 \pm 0.01$ at $(1, 8)$. For $n_\star^v \geq 5$ every row carries a negative gap, from -0.27 ± 0.01 at $(5, 1)$ down to -0.63 ± 0.01 at $(8, 1)$. The shared-sigmoid diagonal cuts straight

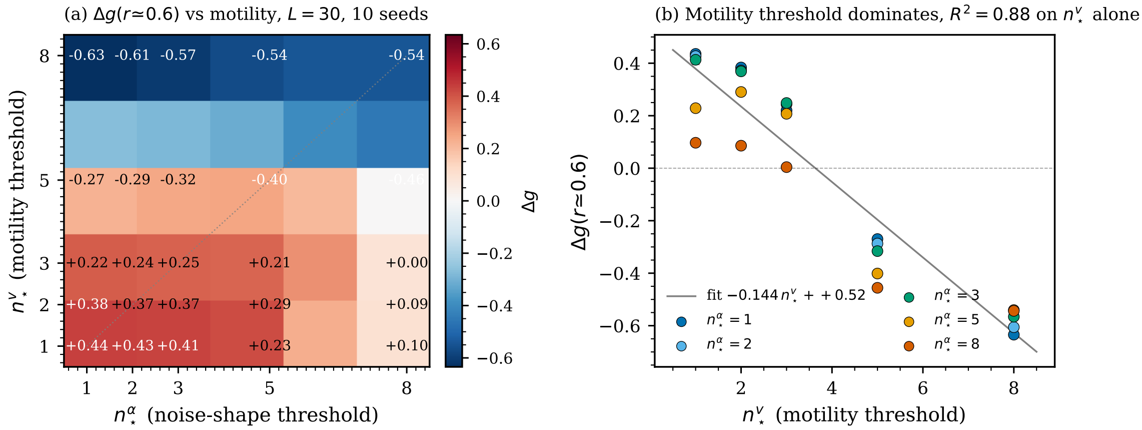

Figure 16. Two-dimensional sweep of the decoupled sigmoid thresholds at $L = 30$, ten seeds per cell. Panel (a): paired gap $\Delta g(r\simeq 0.6) = g_{\rm full}^{\rm decoupled} - g_{\rm motility}$ on the 25-cell grid; the grey dashed line marks the shared-sigmoid locus $n^v_\star = n^\alpha_\star$. Panel (b): the same gap plotted against the motility threshold $n^v_\star$, colour-coded by the noise-shape threshold $n^\alpha_\star$. A linear fit on $n^v_\star$ alone captures $R^2 = 0.88$ of the variance across the grid; $n^\alpha_\star$ alone captures $R^2 = 0.04$.

through the stratification: $(3,3)$ gives $\Delta g = +0.25$ and $(5,5)$ gives $\Delta g = -0.40$, so the sign of the gap is set by $n^v_\star$ alone and not by the timing offset $n^v_\star - n^\alpha_\star$. A linear fit pins this down quantitatively: $\Delta g \simeq -0.144\,n^v_\star + 0.52$ absorbs $R^2 = 0.88$ of the variance, while the matching fit on $n^\alpha_\star$ alone reaches only $R^2 = 0.04$ and the timing-offset fit reaches $R^2 = 0.28$. The motility threshold is, by a clear margin, the dominant control variable.

The mechanism behind this asymmetry is simple. At the shared working point $n_\star = 3$, sitting just above the typical alignment-zone occupancy, a particle inside a dense patch has $n_i > n^v_\star$ and the motility channel engages, while outside the patch v_i saturates at $v_{\max}$. Push $n^v_\star$ to 5 or 8 and the motility channel almost never fires; the decoupled model then collapses onto the noise-shape-only ablation regardless of where $n^\alpha_\star$ sits. That ablation carries a strongly negative dense-quartile correlation $g(r\simeq 0.6) = -0.17$ at the same $L = 30$ (Sec. 3.3.2), so its gap with the motility-only ablation ($g_{\rm dense} = +0.54$) is large and negative — the lower band of the heatmap. Drop $n^v_\star$ back below 3 and the motility channel engages aggressively, dense patches grow, and the noise-shape rectification adds coherence on top of them whether it triggers early (small $n^\alpha_\star$) or late (large $n^\alpha_\star$): the upper band.

The biological reading follows. The density-triggered slowdown is a precondition; without it, the density-dependent reorientation rectification damages dense-phase coherence rather than amplifying it. Heavy-tailed reorientations adapted to local crowding cannot build a flock on their own. They can only sharpen the cohesion that the motility channel has already laid down.

Having located the sign change at one density, we now ask how the critical motility threshold $n^{v,\rm crit}_\star$ moves with the global density σ. The simplest mechanistic reading takes the relevant neighbour count to be the typical alignment-zone occupancy $A\sigma = \pi(R_a^2 - R_r^2)\,\sigma$ and therefore predicts $n^{v,\rm crit}_\star \propto \sigma$ with a positive slope of order $A \simeq 0.75$. We test that directly by sweeping $n^v_\star \in \{1,2,3,4,5,6,8\}$ at three additional densities $\sigma \in \{1.0, 1.5, 3.0\}$ (in addition to $\sigma = 2.22$ already in hand), keeping $n^\alpha_\star = 3$ fixed with ten seeds per cell, and reading off the zero crossing of the per-density linear fit $\Delta g(n^v_\star) \simeq a(\sigma)\,n^v_\star + b(\sigma)$.

Figure 17 collects the result. The per-density fits all reach $R^2 \geq 0.86$, and the critical threshold follows

$$n^{v,\rm crit}_\star(\sigma) = -0.73\,\sigma + 5.81, \qquad R^2 = 0.97, \tag{12}$$

across the four densities $\sigma \in \{1.0, 1.5, 2.22, 3.0\}$. The mean-density prediction is wrong in sign, not just in magnitude: $n^{v,\rm crit}_\star$ *falls* with σ instead of rising with it, and the measured slope -0.73 is the mirror image of the $+0.75$ that $A\sigma$ scaling would have produced.

What drives the sign is the relative density contrast between the dense and dilute phases of the underlying MIPS instability. At low σ the dilute phase is nearly empty, the dense phase carries a large relative-density contrast, and its local neighbour count clears even a moderately demanding $n_\star^v \simeq 5$; the rectification then engages on a sizeable patch and the gap is positive. At high σ the dilute phase is itself crowded, the dense-to-dilute contrast weakens, and $n_\star^v$ has to drop for the rectification to remain a selective dense-phase mechanism. Extrapolating Eq. (12) to $\sigma \simeq 8$ predicts $n_\star^{v,\mathrm{crit}} \to 0$: once the dilute phase is indistinguishable from the dense one, no non-trivial threshold can produce a coherence gain.

Before settling for this density-contrast picture, we subjected it to an obvious falsification check of our own making. If $n_\star^{v,\mathrm{crit}}$ really tracks the dense-phase neighbour count, then it should equal the mean alignment-zone count of dense particles, $\langle n_i \rangle_{\mathrm{dense}}$, measured directly from full-mode trajectories. We measured $\langle n_i \rangle_{\mathrm{dense}}$ on the same four densities (top-density-quartile particles, $R = 1$ local-density radius, $L = 30$, 5 seeds, 12 snapshots) and obtained $\langle n_i \rangle_{\mathrm{dense}} = 2.57, 4.05, 4.85, 5.26$ at $\sigma = 1.0, 1.5, 2.22, 3.0$ — an *increasing* function of σ, running in the opposite direction from $n_\star^{v,\mathrm{crit}}$. The ratio $n_\star^{v,\mathrm{crit}} / \langle n_i \rangle_{\mathrm{dense}}$ moves from 2.08 down to 1.16, 0.85 and 0.69 across the four densities, so the naive equality $n_\star^{v,\mathrm{crit}} = \langle n_i \rangle_{\mathrm{dense}}$ is refuted by our own data. The phenomenology is genuinely more subtle than mean dense-phase density alone. A plausible refinement makes the threshold depend not only on the mean neighbour count inside a patch but also on how long the densest particles spend above it during patch consolidation; we leave a quantitative theory of that transient as an open question.

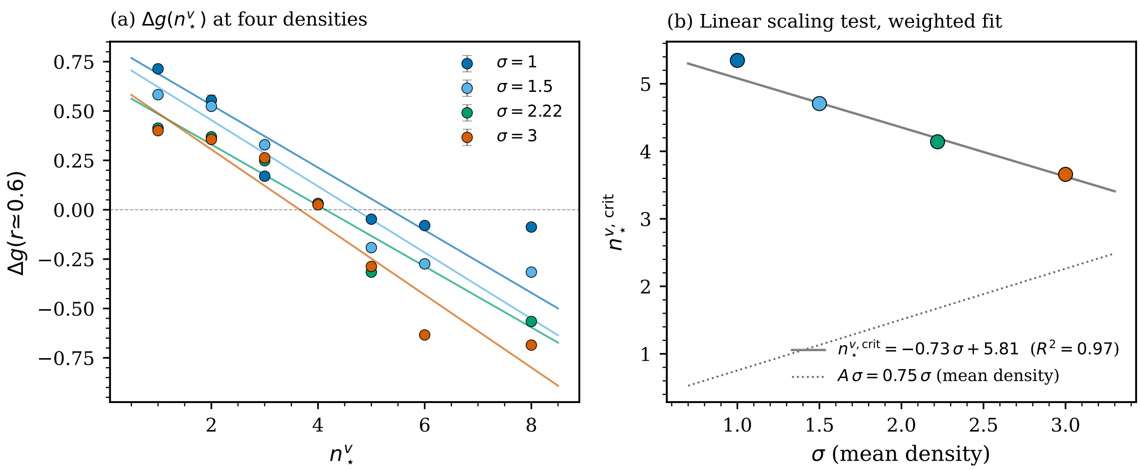

Figure 17. Density-scaling test of the decoupled-sigmoid map. Panel (a): $\Delta g(r \simeq 0.6)$ vs $n_\star^v$ at four global densities $\sigma \in \{1.0, 1.5, 2.22, 3.0\}$ (ten seeds per cell, $n_\star^\alpha = 3$ fixed, $L = 30$), with per-density weighted linear fits. Panel (b): the extracted critical threshold $n_\star^{v,\,\mathrm{crit}}$ at which the gap changes sign, plotted against σ. The weighted linear fit gives $n_\star^{v,\,\mathrm{crit}} = -0.73\,\sigma + 5.81$ with $R^2 = 0.97$; the dotted line is the mean-density prediction $A\sigma$, which would have given a slope of $+0.75$ rather than the observed -0.73.

All of this has been measured under metric alignment, the choice inherited from the original Couzin zonal model, in which every neighbour inside the annulus contributes to the heading update and the alignment-zone count therefore tracks the local density directly. That choice is natural in the metric setting but it conflates two roles: the sigmoid input n_i and the partner set that determines the alignment direction are the same object. So we cannot yet say whether the dominance of $n_\star^v$ belongs to the density-sensing channel (the sigmoid in n_i) or to the density-coupled alignment kernel that the metric rule glues to it. The standard way to pull the two apart is the topological-alignment variant of Ginelli and Chaté [2010]: each particle aligns with its k nearest non-repulsion neighbours, while the sigmoid input n_i is left as the metric annulus count so the density-sensing channel is untouched.

Re-running the 25-cell decoupled-sigmoid map at $L = 30$, $\sigma = 2.22$, with $k = 4$ topological

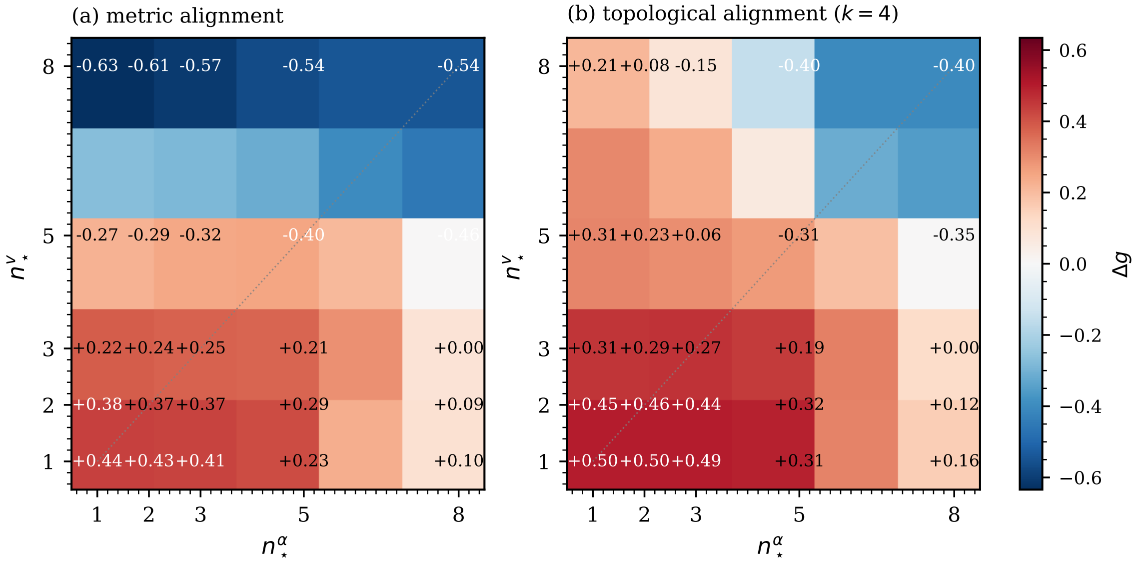

Figure 18. Alignment-kernel comparison of the 2D decoupled-sigmoid map at $L = 30$, $\sigma = 2.22$, ten seeds per cell. Panel (a): metric alignment (same as Fig. 16, reproduced on the matching colour scale). Panel (b): topological alignment with $k = 4$ nearest non-repulsion neighbours. The grey dashed line marks the shared-sigmoid locus $n_\star^v = n_\star^\alpha$.

alignment produces the heatmap of Fig. 18b, shown alongside the metric heatmap of Fig. 16a. The two-band structure is still visible but considerably softened. The linear fit on $n_\star^v$ alone now captures only $R^2 = 0.68$ of the variance (against 0.88 under metric alignment), while the fit on $n_\star^\alpha$ alone climbs to $R^2 = 0.43$ (against 0.04). The bivariate fit $\Delta g \simeq -0.081\, n_\star^v - 0.069\, n_\star^\alpha + 0.73$ assigns the two thresholds nearly equal weight, in sharp contrast with the $-0.144/-0.029$ split of the metric kernel. The sign-flip region retreats as well: cells $(n_\star^v, n_\star^\alpha) = (5, 1)$ and $(8, 1)$, which gave $\Delta g = -0.27$ and -0.63 under metric alignment, now give $+0.31$ and $+0.21$, and only the high-$n_\star^v$, high-$n_\star^\alpha$ corner stays negative.

The motility-threshold dominance is thus a property of the metric-alignment kernel, not a generic feature of the two-feedback dynamics. The reason is geometric. Under metric alignment, the count n_{ali} that feeds the sigmoid is the same count that selects the alignment partners, so pushing the sigmoid threshold above the typical occupancy switches off the slowdown and the neighbour-driven alignment at the same time. Topological alignment breaks that identification: even when $n_\star^v$ sits well above the metric occupancy and the motility channel almost never fires, the $k = 4$ nearest neighbours still hand each particle a baseline alignment, and the noise-shape rectification can sharpen that baseline without dragging the dense-quartile correlation below zero. Biologically, the model speaks most sharply to species that perceive their neighbours on a short metric scale — swarming bacteria, insects in dense plumes — where the slowdown channel does all the work, and less sharply to topological perceivers such as starlings or schooling fish, where the noise-shape threshold pulls roughly its own weight.

One temporal question is left open. The large droplets that the rectification builds in the cluster diagnostic at $L = 30$ are clearly bigger; are they also longer-lived? We tracked the centroids of the connected dense components (same 10×10 threshold finder as in Sec. 3.3.1) over 200 snapshots spaced 50 steps apart, matching them across snapshots by periodic-centroid distance (cutoff 4.0) and particle-count ratio (cutoff 2.0), with ten seeds per mode at $L = 30$ and near-critical η. The mean patch lifetime is 1.60 ± 0.02 snapshots in the motility-only ablation and 1.57 ± 0.02 in the double-adaptive model, a paired-per-seed difference of -0.037 ± 0.029 ($z = -1.30$); the survival distributions $P(\tau' \geq \tau)$ overlap at every lag below $\tau = 10$. The geometric consolidation reported in Sec. 3.3.1 does not buy a longer-lived patch population. The noise-shape rectification reshapes *which* grid cells the dense particles occupy at a given instant, while leaving the turnover of those cells to the motility dynamics alone. It acts on instantaneous geometry, not on temporal persistence.

Taken together, the four tests turn the single-working-point picture of the rectification into something much more constrained. It is a spatial consolidation that needs an aggressive motility threshold and a metric alignment kernel to engage at all; under those conditions it carries on doing its job regardless of where the noise-shape threshold sits, but it never extends the lifetime of the patches it builds, and it switches itself off entirely once the global density washes out the contrast between the dense and dilute phases. The mechanism is local in space and local in time, and it piggybacks on the MIPS instability rather than driving the phase separation on its own.

4 Discussion

4.1 Mechanism and biological interpretation

What ties the two channels together in our setup is the shared sigmoid in n_i: the speed and the stability index move in lockstep. Inside a dense patch the sigmoid pushes $\alpha_i \to \alpha_{\max} = 2$, and the Cauchy bursts that would otherwise scramble a slow particle in the motility-only ablation are replaced by Gaussian-like kicks. Every rectified kick is a heading fluctuation that no longer randomises the local alignment and a polar-order fluctuation that contributes constructively to the coherence the motility channel has begun to build. The noise-shape channel therefore acts as a fluctuation rectifier *inside* the slow region the motility channel carves out, and the two adaptations cooperate where it matters with no operating point at which they fight each other locally. The same mechanism tightens the spatial cohesion of the dense phase: a slow region with rectified noise is denser and more uniform than the same region with Cauchy bursts, so the densest grid cell sits further above the median in the double-adaptive model than in the motility-only ablation, and the spatial cohesion gain shows up clearly at intermediate L with s_{sep} gaps of $+0.24$ at $L = 64$ and $+0.10$ at $L = 90$. The susceptibility synergy $\Delta_n = a_{\text{full}} + a_{\text{Cauchy}} - a_{\text{noise}} - a_{\text{motility}}$ stays positive throughout the seven-size series ($\Delta_4 = +0.69$, $\Delta_5 = +1.45$, $\Delta_6 = +0.46$, $\Delta_7 = +0.48$): both channels feed the same critical scaling, and the full mode scales $\Delta a \simeq +0.6$ steeper than the motility-only ablation, far outside the bootstrap envelope of zero on the four smallest sizes.

The microscopic diagnostics make the picture quantitative. The dense-quartile heading correlation $g(r)$ climbs from 0.542 (motility-only) to 0.695 (double-adaptive) at short range with $z = +14$ at $L = 30$ (Sec. 3.3.2), while the dilute-quartile correlation is essentially flat between the two modes. This dense/dilute asymmetry is the direct fingerprint of the shared sigmoid acting where it is meant to act — swapping Cauchy kicks for Gaussian-like ones inside the slow crowded region, and leaving the dilute phase alone. The density-stratified dense $g(r)$ provides the static face of the signature; its dynamic counterpart, the per-particle dense-phase autocorrelation $C(\tau)$ in Fig. 15, holds a $+0.13$–$+0.24$ gap full versus motility-only over four decades in lag at $L = 30$, with bootstrap 95% CIs that exclude zero throughout. The cluster diagnostic adds the geometric face: the maximum cluster size grows from 55 particles in the motility ablation to 177 particles in the full model ($z = +2.7$), so the noise-shape feedback fuses dense patches into one or two large droplets rather than multiplying smaller ones. The four diagnostics — the FSS synergy on χ, the intermediate-L s_{sep} gap, the dense $g(r)$ gap, and the dense $C(\tau)$ gap — read out the same mechanism on complementary axes (global, spatial, static-local, dynamic-local), and all four are significant at $L = 30$.

The four orthogonal tests of Sec. 3.3.4 draw a sharper portrait than the central working point. The rectification is a spatial-consolidation mechanism, not a temporal-stabilisation one (the patch-lifetime test of Sec. 3.3.4 returns a null result $z = -1.3$); it requires the metric alignment kernel (Sec. 3.3.4: the dominance weakens from $R^2 = 0.88$ to 0.68 under topological alignment); it is controlled by the motility threshold alone under metric alignment (Sec. 3.3.4); and it fails altogether as the dilute phase loses its contrast with the dense one (Sec. 3.3.4: $n_\star^{v,\text{crit}} \to 0$ as $\sigma \to 8$).

A flock that adapts both its motility and its reorientation statistics to local crowding therefore gets the best of both worlds across the sizes we can probe: a spatially separated cohesive phase with reduced internal noise, a steeper susceptibility scaling than either channel on its own, and a robust dense-phase heading correlation. The connection to locust marching [Buhl et al., 2006, Bazazi et al., 2008] and to swarming fish or starling flocks, which operate at group sizes well below the thermodynamic limit, is direct. The two-dimensional decoupled-sigmoid sweep (Sec. 3.3.4) sharpens the biological reading: the gain is governed almost entirely by the motility threshold $n_\star^v$, with the noise-shape threshold $n_\star^\alpha$ a near-negligible second factor under metric alignment (linear regressions: $R^2 = 0.88$ on $n_\star^v$ alone vs $R^2 = 0.04$ on $n_\star^\alpha$ alone). A flock that slows down aggressively at typical local densities reaps the two-feedback cohesion at almost any noise-shape threshold; a flock with an inert motility channel sees the noise-shape adaptation actively damage dense-phase coherence. The biological priority is the density-dependent slowdown, not the order in which the two thresholds are crossed.

4.2 Comparison to the literature

Two strands of the active-matter literature converge on the double-adaptive flock studied here. The motility channel inherits directly from Cates and Tailleur [2015], who proved that a density-dependent speed with $\partial \ln v / \partial \ln \rho < -1$ generates motility-induced phase separation in a non-aligning system. Our motility-only ablation reproduces that prediction inside an aligning Couzin flock: s_{sep} rises from $\simeq 1.2$ for the fixed-speed Cauchy and Vicsek references to $\simeq 1.7$ once the speed sigmoid is switched on, in line with the MIPS fingerprint Cates and Tailleur describe. The result on the soft-tissue side is in the same spirit as Bi et al. [2016], where confluent epithelia jam through a density-driven slowdown; we differ in keeping the alignment interaction, so the dense phase is polar rather than amorphous. Within the metric-Vicsek lineage of Vicsek et al. [1995], the canonical phenomenology is that of Grégoire and Chaté [2004] and Chaté et al. [2008], who established that the metric model with Gaussian noise carries traveling band solitons at finite size and a weakly first-order transition. The band-soliton index $b \leq 1.06$ in Fig. 10 shows that the double-adaptive dynamics does not reach the canonical band regime, and the order-parameter histogram at $L = 30$ stays unimodal with $U_4 \simeq 0.65$ in both motility-only and full modes. The Toner–Tu programme [Toner and Tu, 1995, 1998, Toner, 2012] would predict the same continuous polar order; what it does not anticipate is the dense-quartile correlation gap reported in Fig. 13, because Toner–Tu hydrodynamics treats the noise as a Gaussian field with density-independent variance. Our finding that the susceptibility slope of the full model exceeds that of the motility-only ablation by $\Delta a \simeq +0.6$ adds a quantitative target for a future density-dependent Toner–Tu closure to reproduce.

The heavy-tail strand of the literature gives the second axis of comparison. Reynolds [2018] and Harris et al. [2012] document Lévy-distributed reorientations in foragers and in T-cells respectively, and Ariel et al. [2015] shows that swarming bacteria turn with heavy tails. Our finding on the noise-shape ablation tempers the heavy-tail-as-driver picture: the Cauchy reference and the noise-shape-only ablation both carry susceptibility slopes statistically indistinguishable from zero on the seven-size lever, so heavy-tailed reorientations on their own do *not* drive a critical $\chi_{\max}(L)$ scaling at our density. The alignment averaging in the dense phase washes the bursts out before they propagate. The motility channel is necessary; heavy tails alone are not sufficient. What the noise-shape adaptation contributes is an extra cooperative slope on top of the motility-driven critical scaling — the super-additive $\Delta_n > 0$ trajectory. Couzin et al. [2002] introduced the zonal repulsion-alignment-attraction rule we use, with the original parameters treated as global constants, and we depart from that lineage by making the *stability index* of the noise process a local field on top of the same zonal architecture. Bechinger et al. [2016] review the active-matter zoo within which both feedbacks live, and Marchetti et al. [2013] place them inside a hydrodynamic frame; neither identifies the dense-phase Gaussian conversion we report, because neither couples the noise distribution to the local density. On the experimental side, Korobkova et al. [2004] and Dieterich et al. [2008] establish that bacterial reorientation statistics depend on the local environment, and that is what makes the noise-shape feedback a biologically sensible ingredient rather than a phenomenological convenience.

4.3 Limits and future work

The seven-size FSS series places strong, but not airtight, bounds on the asymptotic synergy. The cumulative Δ_n trajectory ($+0.69, +1.45, +0.46, +0.48$ at $n = 4, 5, 6, 7$) is persistently positive and its 95% bootstrap CI excludes zero at $n = 4$ and $n = 5$. At $n = 6, 7$ the interval widens because the motility slope itself carries a wide bootstrap dispersion ($[+1.60, +3.04]$ on the seven-size fit), but the point estimate stays comfortably above zero. The non-monotonic drop from $\Delta_5 = +1.45$ to $\Delta_6 = +0.46$ tracks the slowing of the double-adaptive $\chi_{\max}$ at $L = 90$, where the largest seed-level fluctuations of the series sit (see Table 1); a higher-statistics re-measurement at $L = 90$ would tell whether Δ_6 stabilises near $+0.5$ or drifts downward toward the asymptotic $a_{\text{full}} - a_{\text{motility}} \simeq +0.6$ implied by the seven-size point estimate. A controlled run at $L = 180$ or $L = 256$ would close the FSS argument by either resolving the residual Δ_n to a finite asymptote or pushing it toward zero.

A handful of natural extensions follow from this picture. Whether $\Delta_7 = +0.48$ persists at larger L is not obvious from the seven sizes alone, and a controlled run at $L = 180$ or $L = 256$ would settle the point; the same run would tell whether the dense-quartile $g(r)$ signal plateaus near its $L = 30$ value of $+0.15$ or erodes with size. Mapping $g(r)$ at intermediate distances $r = 2$ to $r = 5$ would clarify whether the rectification carries a characteristic decay length or is genuinely range-independent within the alignment zone, in the spirit of the heavy-tail correlation analyses of Eliazar and Klafter [2003]. The behavioural plane of Fig. 5 was sampled at $L = 22$ with two seeds for cost reasons; a higher-statistics extension at $L = 30$ and $L = 45$ with five seeds would confirm whether the synergy-positive corner $n_\star \leq 3$, $s \geq 5$ also carries the steepest $\chi_{\max}(L)$ slope. A complementary two-knob sweep over the adaptation amplitudes $v_{\min}/v_{\max}$ and $\alpha_{\min}/\alpha_{\max}$ at fixed $(n_\star, s)$ would isolate the marginal contribution of each channel and let us extrapolate smoothly to the canonical Vicsek limit. The order-parameter histograms at $L \in \{64, 90, 128\}$ stay unimodal across the FSS range (Fig. 9), with mild platykurtosis at $L = 128$ ($\kappa = -0.27$ for motility, -0.20 for full) but no two-peak coexistence; whether $P(\langle\varphi\rangle)$ eventually narrows into a delta peak as $L \to \infty$ or persists as a single broad mode would require an even longer trajectory at $L = 180$ or $L = 256$, beyond the present compute budget. The absence of a metric-Vicsek traveling band, finally, leaves room for a different cohesive structure — a slowly-rotating dense droplet, perhaps — that a closer real-space analysis at sizes beyond $L = 30$ should reveal.

Two model variants offer obvious robustness tests. A topological alignment rule, in which each particle aligns with its k nearest neighbours rather than with the metric annulus, has been shown to flip the heavy-tail trend in the canonical Vicsek–Couzin family, and combining the topological rule with our two-feedback dynamics is a one-line edit of the interaction kernel. The two-dimensional decoupled-sigmoid map of Sec. 3.3.4 already shows that the motility threshold $n_\star^v$ is the single dominant control variable on a 25-cell grid, and the σ-sweep of Sec. 3.3.4 establishes an empirical linear law $n_\star^{v,\text{crit}}(\sigma) = -0.73\,\sigma + 5.81$ with $R^2 = 0.97$ that runs opposite to the mean-density prediction $A\sigma$. A hydrodynamic theory that recovers both the linear $\Delta g(n_\star^v)$ dependence at fixed density and the negative slope of the σ scaling — attributable to the MIPS dense-vs-dilute density contrast rather than the mean density — is a natural extension. Beyond the microscopic level, a hydrodynamic description in the Toner–Tu spirit [Toner and Tu, 1995, Toner, 2012] would put the mechanism on a quantitative footing. The natural slow fields are $\rho(\mathbf{x}, t)$ and $\mathbf{p}(\mathbf{x}, t)$, with an effective stability index $\alpha_{\text{eff}}(\rho)$ and an effective speed $v_{\text{eff}}(\rho)$ inherited from the shared sigmoid; in that frame the Cates–Tailleur criterion $\partial \ln v/\partial \ln \rho < -1$ generalises naturally, and the heavy-tail correction to the noise variance $\propto \eta^{2/\alpha_{\text{eff}}(\rho)}$ shifts the spinodal boundary. We plan to extract $\alpha_{\text{eff}}(\rho)$ and $v_{\text{eff}}(\rho)$ from the simulation directly and to fit a Toner–Tu form to the deeply ordered regime, so the effective γ/ν of the double-adaptive transition can be benchmarked against the metric Vicsek prediction. A movie utility shipped with the code writes MP4 sequences for each of the five modes at chosen η values, with each frame coloured by v_i and an inset tracking $\langle\varphi\rangle(t)$ and $s_{\text{sep}}(t)$, so that the rectification is visible in real time; the movies are part of the supplementary material and complement the static figures of this paper.

A Hydrodynamic description and the asymmetry of decoupled sigmoids

The microscopic model of Sec. 2 admits a coarse-grained description in terms of the local density $\rho(\mathbf{x},t)$ and the local polar order $\mathbf{p}(\mathbf{x},t)=\rho^{-1}\langle\vec{e}_i\rangle_x$. We sketch the derivation here at the level of physical arguments rather than a full Boltzmann coarse-graining, the goal being to make the rectification mechanism and the decoupled-sigmoid asymmetry of Sec. 3.3.4 analytically transparent.

A.1 Continuum fields and conservation

The instantaneous neighbour count of a particle is, in the continuum limit,

$$n_i \approx A\,\rho(\mathbf{x}_i), \qquad A \equiv \pi(R_a^2 - R_r^2), \tag{13}$$

where $A \simeq 0.75$ for the standard parameters $R_a = 0.7$ and $R_r = 0.5$. Inserting Eq. (13) into Eqs. (4)–(6) replaces the per-particle adaptations by smooth density-dependent fields,

$$\sigma(\rho) = \frac{1}{1+\exp\!\big[-s\,(A\rho - n_\star)\big]}, \tag{14}$$

$$v(\rho) = v_{\max} - \Delta v\,\sigma(\rho), \quad \Delta v \equiv v_{\max} - v_{\min}, \tag{15}$$

$$\alpha(\rho) = \alpha_{\min} + \Delta\alpha\,\sigma(\rho), \quad \Delta\alpha \equiv \alpha_{\max} - \alpha_{\min}. \tag{16}$$

Density conservation gives the standard active-matter form

$$\partial_t\rho + \nabla\cdot\big[\rho\,v(\rho)\,\mathbf{p}\big] = 0, \tag{17}$$

while the polar order field obeys, to leading order in gradients and $|\mathbf{p}|$,

$$\partial_t\mathbf{p} = -\Gamma(\rho,\eta)\,\mathbf{p} + \nu\,\nabla^2\mathbf{p} - \lambda\,(\mathbf{p}\cdot\nabla)\mathbf{p} + \cdots, \tag{18}$$

where the first term is the rotational decorrelation rate set by the angular noise and ν, λ are the standard Toner–Tu coefficients [Toner and Tu, 1995, Toner, 2012]. The non-trivial input from our model is that Γ depends on ρ through the local stability index $\alpha(\rho)$.

A.2 Effective rotational decorrelation

A symmetric α-stable angular kick $\xi \sim S_\alpha(c{=}\eta)$ has characteristic function $\langle e^{ik\xi}\rangle = \exp(-\eta^\alpha|k|^\alpha)$, so the single-step decorrelation of an aligned heading is $\langle\cos\xi\rangle = \exp(-\eta^\alpha)$ and, for small η, the polar-order decay rate per step reads

$$\Gamma(\rho,\eta) = 1 - \langle\cos\xi\rangle \;\simeq\; \eta^{\alpha(\rho)}. \tag{19}$$

The decorrelation rate is therefore $\eta^{\alpha_{\min}} = \eta$ in the dilute limit (Cauchy noise) and $\eta^{\alpha_{\max}} = \eta^2$ in the dense limit (Gaussian noise). For $\eta < 1$ the dense-phase decorrelation is smaller than the dilute-phase one by a factor η, about ten at the working point $\eta = 0.1$,

$$\frac{\Gamma_{\text{dense}}}{\Gamma_{\text{dilute}}} = \eta^{\,\alpha_{\max}-\alpha_{\min}} = \eta, \tag{20}$$

the analytic counterpart of the rectification observed in the simulation: the noise-shape feedback converts the η-linear Cauchy decorrelation into the much weaker η^2 Gaussian one inside the dense phase.

A.3 MIPS instability and its rectification correction

A linear-stability analysis of the homogeneous polar state $\rho = \rho_0$, $\mathbf{p} = p_0\hat{\mathbf{e}}$ under a longitudinal density perturbation $\delta\rho \propto e^{ikx+\Omega t}$ couples to a longitudinal polar perturbation $\delta p_\parallel$. The density branch develops the Cates–Tailleur instability [Cates and Tailleur, 2015] when the effective compressibility goes negative,

$$\left.\frac{\partial \ln v(\rho)}{\partial \ln \rho}\right|_{\rho_0} < -1. \tag{21}$$

For the sigmoidal $v(\rho)$ of Eq. (15) this criterion is met in a window of densities centred on $A\rho_0 \simeq n_\star$, of width set by $1/s$. The polar branch adds a term proportional to $\Gamma(\rho_0, \eta) = \eta^{\alpha(\rho_0)}$. In the shared-sigmoid model $v(\rho)$ and $\alpha(\rho)$ vary together, so the instability window of Eq. (21) sits at the same density at which the rectification $\Gamma \to \eta^2$ kicks in: the dense phase that nucleates there is born already rectified, and the inner Γ is small from the moment it forms.

A.4 Decoupled sigmoids: a sequential argument and what it does not capture

The decoupled-sigmoid extension of Sec. 3.3.4 replaces Eq. (14) by two independent profiles $\sigma_v(\rho)$ and $\sigma_\alpha(\rho)$, with thresholds $n_\star^v$ and $n_\star^\alpha$ respectively. The MIPS instability of Eq. (21) is then controlled by $\sigma_v(\rho)$ alone, while the rectification $\Gamma \sim \eta^{\alpha(\rho)}$ is controlled by $\sigma_\alpha(\rho)$ alone. A sequential argument along the $n_\star^v - n_\star^\alpha$ axis is instructive but incomplete; we work it through here and then explain why it falls short of the empirical map.

When $n_\star^v < n_\star^\alpha$ (the motility-first quadrant), the MIPS instability triggers first at a density $A\rho_v \simeq n_\star^v$, where the noise is still Cauchy since $A\rho_v \ll n_\star^\alpha$ implies $\alpha \simeq \alpha_{\min}$. The dense phase grows to a density $A\rho_{\text{dense}} \gtrsim n_\star^\alpha$ at which the rectification turns on, so the rectification acts on a fully formed slow region. The inner Γ drops from η to η^2 *after* the patch has consolidated, locking in the coherence; the dense-quartile $g(r)$ enhancement is predicted to be large and positive.

When $n_\star^v > n_\star^\alpha$ (the alpha-first quadrant), the rectification triggers first at $A\rho_\alpha \simeq n_\star^\alpha$, where particles still move at $v \simeq v_{\max}$. Fast Gaussian-like particles do not preferentially accumulate, but they experience the alignment sum on a dilute background of Cauchy-noisy neighbours: the locally Gaussian particle takes the Cauchy mean of the surrounding kicks at full force and is deflected harder than a uniformly Cauchy particle would be. The dense-quartile correlation is predicted to be damaged rather than enhanced.

This sequential picture captures the sign of Δg along the diagonal $n_\star^v - n_\star^\alpha$ in the right direction, but it misses what the empirical map of Sec. 3.3.4 actually shows. The two-band structure in Fig. 16 runs along horizontal lines of constant $n_\star^v$, not along diagonals of constant timing difference: the linear fit on $n_\star^v$ alone captures $R^2 = 0.88$ of the variance, while the fit on $n_\star^v - n_\star^\alpha$ reaches only $R^2 = 0.28$. The argument above accounts for the mode-ordering motility-first $\succ$ shared $\succ$ alpha-first along the diagonal, but it does not explain why, e.g., $(n_\star^v, n_\star^\alpha) = (5, 8)$ (a motility-first cell with both thresholds high) is still firmly in the negative band. The right control variable is the engagement of the motility channel at typical local densities, set by $n_\star^v$ relative to σ, not by its difference with $n_\star^\alpha$; a Boltzmann-style coarse-graining that tracks the full patch-formation dynamics rather than the simple instability-window argument would be needed to recover the $n_\star^v$-dominance quantitatively. We leave that derivation as open.

A.5 Toner–Tu interpretation

In the language of Toner–Tu hydrodynamics with density-dependent coefficients, our two-feedback model maps onto a flock with $v_{\text{eff}}(\rho)$ and a fractional rotational diffusion $D_{\text{rot}}(\rho) \propto \eta^{\alpha(\rho)}$. The fractional dependence on η is the imprint of the non-Gaussian angular noise, and it turns the density-dependent stability index into a non-trivial ρ-dependent damping of the polar-order field. Reproducing the $n_\star^v$-dominance reported in Sec. 3.3.4 from Eqs. (17) and (18) requires identifying the critical motility threshold at which the dense-phase fraction collapses and propagating its effect through the noise-shape sector. The linear-stability sketch above captures the qualitative two-band structure on the diagonal but neither the empirical horizontal-band structure of the $(n^v_\star, n^\alpha_\star)$ map nor the σ-scaling law $n^{v,\mathrm{crit}}_\star(\sigma) = -0.73\,\sigma + 5.81$, which runs opposite to the mean-density prediction $A\sigma$ (Sec. 3.3.4). The motility channel must engage at typical *dense-phase* local densities for the rectification to add coherence to the patch it builds; whether those densities scale with σ as the MIPS dense-to-dilute contrast scales remains the natural direction for a quantitative theory.

Code and data availability

The simulation code, the four-mode driver, the small-η robustness check, the $(n_\star, s)$ plane sweep, the real-space snapshot, the order-parameter histograms, the polar-axis profile, the Vicsek–Gaussian reference run, the movie-generation utility and the figure-generation scripts are released as supplementary material together with the data. All runs reported here used Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1, Numba 0.65.1 (with LLVM 0.47.0) and Matplotlib 3.10.9 on Linux x86_64. The interaction loop is JIT-compiled with Numba's parallel reduction; per-particle random streams are derived from a single NumPy `Generator` with a per-run seed, so each `(seed, mode,` L`)` tuple is reproducible bit-for-bit on a given architecture.

References

G. Ariel, A. Rabani, S. Benisty, J. D. Partridge, R. M. Harshey, and A. Be'er. Swarming bacteria migrate by lévy walk. *Nature Communications*, 6:8396, 2015. doi: 10.1038/ncomms9396.

S. Bazazi, J. Buhl, J. J. Hale, M. L. Anstey, G. A. Sword, S. J. Simpson, and I. D. Couzin. Collective motion and cannibalism in locust migratory bands. *Current Biology*, 18:735–739, 2008. doi: 10.1016/j.cub.2008.04.035.

C. Bechinger, R. Di Leonardo, H. Löwen, C. Reichhardt, G. Volpe, and G. Volpe. Active particles in complex and crowded environments. *Rev. Mod. Phys.*, 88:045006, 2016. doi: 10.1103/RevModPhys.88.045006.

D. Bi, X. Yang, M. C. Marchetti, and M. L. Manning. Motility-driven glass and jamming transitions in biological tissues. *Phys. Rev. X*, 6:021011, 2016. doi: 10.1103/PhysRevX.6.021011.

J. Buhl, D. J. T. Sumpter, I. D. Couzin, J. J. Hale, E. Despland, E. R. Miller, and S. J. Simpson. From disorder to order in marching locusts. *Science*, 312:1402–1406, 2006. doi: 10.1126/science.1125142.

M. E. Cates and J. Tailleur. Motility-induced phase separation. *Annu. Rev. Condens. Matter Phys.*, 6:219–244, 2015. doi: 10.1146/annurev-conmatphys-031214-014710.

J. M. Chambers, C. L. Mallows, and B. W. Stuck. A method for simulating stable random variables. *J. Amer. Statist. Assoc.*, 71:340–344, 1976. doi: 10.1080/01621459.1976.10480344.

H. Chaté, F. Ginelli, G. Grégoire, and F. Raynaud. Collective motion of self-propelled particles interacting without cohesion. *Phys. Rev. E*, 77:046113, 2008. doi: 10.1103/PhysRevE.77.046113.

I. D. Couzin, J. Krause, R. James, G. D. Ruxton, and N. R. Franks. Collective memory and spatial sorting in animal groups. *J. Theor. Biol.*, 218:1–11, 2002. doi: 10.1006/jtbi.2002.3065.

P. Dieterich, R. Klages, R. Preuss, and A. Schwab. Anomalous dynamics of cell migration. *Proc. Natl. Acad. Sci. USA*, 105:459–463, 2008. doi: 10.1073/pnas.0707603105.

I. Eliazar and J. Klafter. Lévy-driven langevin systems: Targeted stochasticity. *J. Stat. Phys.*, 111:739–768, 2003. doi: 10.1023/A:1022894030773.

F. Ginelli and H. Chaté. Relevance of metric-free interactions in flocking phenomena. *Phys. Rev. Lett.*, 105:168103, 2010. doi: 10.1103/PhysRevLett.105.168103.

B. D. Greenshields. A study of traffic capacity. *Highway Research Board Proc.*, 14:448–477, 1935.

G. Grégoire and H. Chaté. Onset of collective and cohesive motion. *Phys. Rev. Lett.*, 92:025702, 2004. doi: 10.1103/PhysRevLett.92.025702.

T. H. Harris, E. J. Banigan, D. A. Christian, C. Konradt, E. D. T. Wojno, K. Norose, E. H. Wilson, B. John, W. Weninger, A. D. Luster, A. J. Liu, and C. A. Hunter. Generalized Lévy walks and the role of chemokines in migration of effector $CD8^{+}$ T cells. *Nature*, 486:545–548, 2012. doi: 10.1038/nature11098.

E. Korobkova, T. Emonet, J. M. G. Vilar, T. S. Shimizu, and P. Cluzel. From molecular noise to behavioural variability in a single bacterium. *Nature*, 428:574–578, 2004. doi: 10.1038/nature02404.

M. J. Lighthill and G. B. Whitham. On kinematic waves ii: A theory of traffic flow on long crowded roads. *Proc. R. Soc. Lond. A*, 229:317–345, 1955. doi: 10.1098/rspa.1955.0089.

M. C. Marchetti, J.-F. Joanny, S. Ramaswamy, T. B. Liverpool, J. Prost, M. Rao, and R. A. Simha. Hydrodynamics of soft active matter. *Rev. Mod. Phys.*, 85:1143–1189, 2013. doi: 10.1103/RevModPhys.85.1143.

A. M. Reynolds. Current status and future directions of Lévy walk research. *Biology Open*, 7: bio030106, 2018. doi: 10.1242/bio.030106.

J. Toner. Reanalysis of the hydrodynamic theory of fluid, polar-ordered flocks. *Phys. Rev. E*, 86: 031918, 2012. doi: 10.1103/PhysRevE.86.031918.

J. Toner and Y. Tu. Long-range order in a two-dimensional dynamical XY model: How birds fly together. *Phys. Rev. Lett.*, 75:4326–4329, 1995. doi: 10.1103/PhysRevLett.75.4326.

J. Toner and Y. Tu. Flocks, herds, and schools: A quantitative theory of flocking. *Phys. Rev. E*, 58:4828, 1998. doi: 10.1103/PhysRevE.58.4828.

T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Shochet. Novel type of phase transition in a system of self-driven particles. *Phys. Rev. Lett.*, 75:1226–1229, 1995. doi: 10.1103/PhysRevLett.75.1226.